

qt-Pegasis: Simpler and Faster Effective Class Group Actions

Pierrick Dartois^{1,2,3}, Jonathan Komada Eriksen⁴, Riccardo Invernizzi⁴ and Frederik Vercauteren⁴

¹ Univ. Rennes, Inria, CNRS, IRISA, UMR 6074, F-35000, Rennes, France

² Univ. Bordeaux, CNRS, Bordeaux INP, IMB, UMR 5251, F-33400, Talence, France

³ INRIA, IMB, UMR 5251, F-33400, Talence, France

⁴ KU Leuven, COSIC, Heverlee, Belgium

Abstract. In this paper, we revisit the recent **Pegasis** algorithm that computes an effective group action of the class group of any imaginary quadratic order R on a set of supersingular elliptic curves primitively oriented by R . Although **Pegasis** was the first algorithm showing the practicality of computing unrestricted class group actions at higher security levels, it is complicated and prone to failures, which leads to many rerandomizations.

We present a new algorithm, **qt-Pegasis**, which is much simpler, but at the same time faster and removes the need for rerandomization of the ideal we want to act with, since it never fails. It leverages the main technique of the recent **Qlapoti** approach. However, **Qlapoti** solves a norm equation in a quaternion algebra, which corresponds to the full endomorphism ring of a supersingular elliptic curve. We show that the algorithm still applies in the quadratic setting, by embedding the quadratic ideal into a quaternion ideal using a technique similar to the one applied in **KLaPoTi**. This way, we can reinterpret the output of **Qlapoti** as four equivalent quadratic ideals, instead of two equivalent quaternion ideals. We then show how to construct a **Clapoti**-like diagram in dimension 2, which embeds the action of the ideal in a 4-dimensional isogeny.

We implemented our **qt-Pegasis** algorithm in SageMath for the CSURF group action, and we achieve a speedup over **Pegasis** of $1.8\times$ for the 500-bit parameters and $2.6\times$ for the 4000-bit parameters.

1 Introduction

An essential algorithmic subroutine in isogeny-based cryptography is to translate ideals of (a subring of) the endomorphism ring of an elliptic curve E , to its corresponding isogeny. For instance, when $\text{End}(E)$ is a maximal quaternion order, converting $\text{End}(E)$ -ideals to their isogenies forms the basis of the signing procedures of both **SQLsign** [1] and **PRISM** [4], while translating ideals of a quadratic subring of $\text{End}(E)$ gives the only known way to construct commutative group-action based primitives [20, 47, 14, 13, 26, 24].

In both the quaternionic and the quadratic case, the degree of the corresponding isogeny equals the reduced norm of the given ideal I . The fastest known

algorithms to compute an isogeny directly from the ideal I are still exponential in $\log \ell$ [48, 7], where ℓ is the largest prime dividing the degree of the isogeny. The more efficient methods all exploit the fact that an ideal J equivalent to I results in the same co-domain curve (up to isomorphism). As such we are free to replace I by any equivalent ideal J that has nicer properties, e.g. an equivalent ideal $J \sim I$ satisfying $\text{nrd}(J) = M$, for some smooth number M , which makes computing the isogeny corresponding to J (and hence I) feasible.

In the quaternionic case, the above norm equation is partially solved by the KLPT algorithm [33], which, among many other things, was used to construct the first versions of **SQlsign** [27, 28, 18]. However, given a quaternion ideal I , the KLPT algorithm is only able to find $J \sim I$ of norm $M > p^3$. The size of this output leads to big complications in practice, and for **SQlsign**, it meant that one had to apply a costly procedure involving many intermediate steps. Further, the output distribution of KLPT is not well understood, which led to the first versions of **SQlsign** being based on very ad-hoc hardness assumptions.

In the quadratic case, the situation is even worse, as there are no known polynomial-time algorithms to find equivalent quadratic ideals of smooth norm. In CSIDH [14], this is manifested by the fact that one a priori only samples ideals of smooth norm, resulting in a restricted effective group action. While this is sufficient to construct a non-interactive key-exchange, more complex protocols become insecure without further tools in this setting [38, 29, 36]. In CSI-FiSh [9], it was shown how a superpolynomial-time precomputation can turn CSIDH into an unrestricted effective group action. Further work on scaling this construction led to SCALLOP [26] and related primitives [15, 3], however neither of these take care of the superpolynomial-time precomputation, which means that SCALLOP-based primitives were only able to reach security levels equivalent to CSIDH-1500, while optimistic estimates suggest that CSIDH-2000 is the minimum for post-quantum security equivalent to NIST level 1 [8, 11, 44].

However, both the quadratic and the quaternionic case have recently been massively improved. The starting point was when Robert and Page introduced Clapoti [42], which showed how one could translate an ideal I by instead looking for equivalent ideals $J_1, J_2 \sim I$, solving the significantly easier norm equation

$$u \cdot \text{nrd}(J_1) + v \cdot \text{nrd}(J_2) = M, \quad (1)$$

by embedding the corresponding isogeny in an isogeny of higher-dimensional abelian varieties. In the quaternionic setting, this both created the new signature scheme **PRISM** [4], and in **SQlsign** this gave huge improvements to the signing procedure [5], while in the quadratic setting, Clapoti itself marked the first polynomial time algorithm to evaluate the action of arbitrary elements of the class group, though it was mainly a theoretical construction.

However, the original algorithm to solve Equation (1) fails with non-negligible probability when applied to $M < p$, where $-p$ is the discriminant of the relevant order. In the quaternionic case, this mainly led to a complicated rerandomization procedure for **PRISM** and **SQlsign**, and for **SQlsign** specifically, it even created an issue in the security proof [2]. In the quadratic case, the situation was even

worse, as this algorithm did not work at all when applied directly to the CSIDH orientation, and $M < p$, as there is the additional requirement that u and v must be written as sums of squares if one wants to stay in dimension 4. However, by factoring out smooth parts of $u, v, n(J_1)$ and $n(J_2)$, it becomes possible, as shown with the introduction of **Pegasis** [25], which gave the first efficient way of evaluating the action of arbitrary ideals in the CSIDH setting.

In the quaternionic setting, the issue of auxiliary isogenies of degrees u and v , was taken care of in **Qlapoti** [12], where it was shown how to solve the less general norm equation

$$\text{nrd}(J_1) + \text{nrd}(J_2) = M$$

directly, with an algorithm that fails with only negligible probability, and which leads to a significantly more efficient procedure for translating I to the corresponding isogeny, due to the absence of u and v . For **PRISM** and **SQIsign**, this again led to significant improvements in efficiency and big simplifications of the relevant algorithms, while for **SQIsign** specifically, it also closed the final gap in the security proof.

Our contribution. We show how to apply **Qlapoti** to **Pegasis**. Even though **Qlapoti** is specifically an algorithm for solving a norm equation of quaternion ideals, we show how to apply it to the quadratic setting, by embedding the quadratic ideal in a quaternion ideal. As shown in **KLaPoTi**, the output of **KLPT** (which is a quaternion ideal of given norm) can be reinterpreted as two equivalent quadratic ideals, whose norm sum to the given norm. In the same way, the output of **Qlapoti**, which are two equivalent quaternion ideals whose norm sum to a given number M , can be reinterpreted as four equivalent quadratic ideals whose norms also sum to M . To turn this output into something useful, we show that applying the technique from **Clapoti** twice, gives an isogeny in dimension 4 of degree M , which embeds the action of the ideals in question.

Since **Qlapoti** is significantly more reliable than the algorithm for solving the norm-equation on which **Pegasis** was based, we avoid having to factor out smooth parts of the norm equation, which avoids using Elkies algorithm [31] entirely. Further, we note that the fact that the ideal comes from a quadratic ideal gives it a lot of extra structure. This allows us to create a tailored version of **Qlapoti**, which is even faster than the general algorithm, and avoids lattice-reduction entirely, which makes constant-time implementations less daunting.

Our implementation shows that the resulting variant of **Pegasis**, called **qt-Pegasis**, is between $1.8\times$ and $2.6\times$ faster depending on the security level. Further, essentially the whole cost of evaluating the action by an element of the class group comes from computing a single chain of 2-isogenies in dimension 4 in **qt-Pegasis**; in our implementation this takes up between 91% and 93% of the total time. Our implementation is available at

<https://github.com/KULeuven-COSIC/qt-pegasis>

Technical Overview. Let $R = \mathbb{Z}[\omega]$ be an arbitrary imaginary quadratic order of discriminant Δ_R . Let $K = R \otimes \mathbb{Q}$, and consider the quaternion algebra $B = K + \mathbf{i}K$, with multiplication laws defined by $\mathbf{i}^2 = -1$, and $\mathbf{i}\omega = \bar{\omega}\mathbf{i}$. Note that B is in general different from $B_{p,\infty}$ since ramification at p requires $p \mid \Delta_K$. For CSURF, we have $\omega^2 = -p$ with $p \equiv 3 \pmod{4}$ and thus $B = B_{p,\infty}$, the quaternion algebra ramified at p (where $\mathbb{F}_p$ is the field of definition of the curves). We want to emphasize that our algorithm works for any order R , and not just for the CSURF order; the only requirement is that $M = 2^e \approx \Delta_R$ (and, in practice, that the 2^e torsion is readily available on E).

Given an ideal $I \subseteq \mathcal{O}_0$, where $\mathcal{O}_0 = R + \mathbf{i}R \subseteq B$, **Qlapoti** is an algorithm for finding two elements $\alpha_1, \alpha_2 \in I$ satisfying

$$\text{nrd}(\alpha_1) + \text{nrd}(\alpha_2) = 2^e \cdot \text{nrd}(I),$$

where 2^e is close to Δ_R . When $\Delta_R = p$ is a prime $p \equiv 3 \pmod{4}$, $\mathcal{O}_0$ is equal to the endomorphism ring of the supersingular curve $E_0/\mathbb{F}_p$ with $j(E_0) = 1728$, and **Qlapoti** thus gives a simple and efficient way of translating I to its corresponding isogeny $\phi_I : E_0 \rightarrow E_I$ (under the Deuring correspondence), by embedding it in a 2-dimensional 2^e -isogeny, as shown in **Clapoti**.

The setting for this work however, is as follows: given any elliptic curve E with an orientation by R , and an ideal $\mathfrak{a} \subseteq R$, we are interested in computing the class group action $\mathfrak{a} \star E$, i.e. the corresponding isogeny $\phi_{\mathfrak{a}} : E \rightarrow E_{\mathfrak{a}}$. Consider the quaternion ideal $\mathfrak{a} + \mathbf{i}\mathfrak{a} \subseteq \mathcal{O}_0$. Applying **Qlapoti** to this ideal gives the elements $\alpha_1, \alpha_2 \in \mathfrak{a} + \mathbf{i}\mathfrak{a}$. Rewriting $\alpha_i = \beta_i + \mathbf{i}\gamma_i$, with $\beta_i, \gamma_i \in \mathfrak{a}$, we see that **Qlapoti** actually gives a solution

$$\text{n}(\beta_1) + \text{n}(\beta_2) + \text{n}(\gamma_1) + \text{n}(\gamma_2) = 2^e \cdot \text{n}(\mathfrak{a}). \quad (2)$$

Now, these four elements will give us a 2^e -isogeny, this time in dimension 4, embedding our isogeny $\phi_{\mathfrak{a}}$. We do this by first applying **Clapoti** to β_i, γ_i for $i = 1, 2$ separately, resulting in two isogenies Φ_i in dimension 2, with the same domain and codomain, whose degree sum to 2^e . Another simple application of Kani's lemma to the commutative diagram

$$\begin{array}{ccc} A & \longrightarrow & E \times E \\ \uparrow & & \uparrow \tilde{\Phi}_2 \\ E \times E & \xrightarrow{\Phi_1} & E_{\mathfrak{a}} \times E_{\bar{\mathfrak{a}}} \end{array}$$

then gives the desired isogeny in dimension 4; see Proposition 2 for details.

Although the approach outlined above already works, we can do better: First, we take advantage of the special structure of the ideal $\mathfrak{a} + \mathbf{i}\mathfrak{a}$, to create a new variant of **Qlapoti**. Both variants require rerandomizing until a certain 2-dimensional CVP instance has a particularly close solution. However, by taking advantage of the structure of the ideal, we may set up this CVP instance in a very predictable way, allowing us to see if the solution is good enough before applying Babai rounding to actually compute it.

Once we have found a solution to Equation (2), we have to compute the 4-dimensional 2^e -isogeny $F : E^4 \rightarrow E_{\mathfrak{a}} \times E_{\bar{\mathfrak{a}}} \times A$ obtained by Kani's lemma. We compute it as a chain of 2-isogenies using level 2 theta coordinates (as we shall explain further in Section 2.4). We give more details on the computation of F in Section 4, in particular on how to compute theta coordinates on E^4 that are adapted to F . In this section, we also study some singularities that can appear at the first steps of the chain (*e.g.* endomorphisms, delayed gluing or zero theta constants) that may complicate this computation. We explain how to detect and avoid these singularities beforehand.

Acknowledgements. This work is supported by the European Research Council (ERC) under the European Union's Horizon 2020 research and innovation programme (grant agreement ISOCRYPT – No. 101020788), by the Research Council KU Leuven grant C14/24/099 and by CyberSecurity Research Flanders with reference number VOEWICS02. This research has also been supported by the France 2030 program under grants ANR-22-PETQ-0008 (PQ-TLS) and BPI France project RESQUE. R. Invernizzi is funded by Research Foundation - Flanders (FWO) under a PhD Fellowship fundamental research (project number 1138925N). We thank Steven Galbraith and the anonymous reviewers for helpful feedback.

2 Preliminaries

We let $K = \mathbb{Q}(\sqrt{d})$ with $d < 0$ a square-free integer denote an imaginary quadratic number field with ring of integers $R_K = \mathbb{Z}[\omega_K]$, where $\omega_K = \frac{1+\sqrt{d}}{2}$ or $\omega_K = \sqrt{d}$ depending on whether $d \equiv 1 \pmod{4}$ or not. Let $\Delta_K = \text{tr}(\omega_K)^2 - 4\text{n}(\omega_K)$ denote the discriminant of R_K . For an order $R \subset R_K$ we denote $f = [R_K : R]$ its conductor and we can write $R = \mathbb{Z}[\omega_R]$ where $\omega_R = f\omega_K$ and $\Delta_R = f^2\Delta_K$. The class group $\text{Cl}(R)$ consists of invertible fractional R -ideals modulo principal ideals. Furthermore, by Minkowski, every ideal class $[\mathfrak{a}] \in \text{Cl}(R)$ contains an ideal $\mathfrak{a}$ with $\text{n}(\mathfrak{a}) < \frac{2}{\pi}\sqrt{-\Delta_R}$.

For a general introduction to isogenies in cryptographic contexts, see [32].

2.1 The class group action on oriented curves

The class group $\text{Cl}(R)$ of an imaginary quadratic order R acts freely and transitively on the set $\text{Ord}_q(R)$ of ordinary elliptic curves $E/\mathbb{F}_q$ with $\text{End}(E) \cong R$ [49, Th. 4.5]. Colò and Kohel [17] introduced an analogous action for supersingular elliptic curves, that was priorly known to Belding [6] in another language. We summarize this theory below along with results from Onuki [41].

Let $R = \mathbb{Z}[\alpha]$ be an imaginary quadratic order and $E/\mathbb{F}_{p^n}$ an elliptic curve. An injective ring morphism $\iota : R \hookrightarrow \text{End}(E)$ is called an *R -orientation*; it is *primitive* if ι cannot be extended to an embedding $R' \hookrightarrow \text{End}(E)$ for $R \subsetneq R'$. The pair (E, ι) is called a (primitively) *R -oriented elliptic curve*.

Given a primitively R -oriented (E, ι) and an integral invertible ideal $\mathfrak{a} \subseteq R$ with norm prime to p , let $\varphi_{\mathfrak{a}}: E \rightarrow E_{\mathfrak{a}}$ be the isogeny with kernel

$$E[\mathfrak{a}] = \bigcap_{\sigma \in \mathfrak{a}} \ker(\iota(\sigma))$$

of degree $n(\mathfrak{a})$. The induced map $R \rightarrow \text{End}(E_{\mathfrak{a}}); \gamma \mapsto \varphi_{\mathfrak{a}}\iota(\gamma)\widehat{\varphi}_{\mathfrak{a}}$ becomes a ring morphism after normalization, yielding a primitive R -orientation $(\varphi_{\mathfrak{a}})_*(\iota)$ on $E_{\mathfrak{a}}$ [41, Prop. 3.5].

The set $\text{SS}_{p^n}^{\text{pr}}(R)$ of R -isomorphism classes of primitively R -oriented supersingular curves over $\mathbb{F}_{p^n}$ is non-empty precisely when p is non-split in $K = R \otimes_{\mathbb{Z}} \mathbb{Q}$ and does not divide R 's conductor in R_K [41, Prop. 3.2]. The action

$$\begin{aligned} \text{Cl}(R) \times \text{SS}_p^{\text{pr}}(R) &\rightarrow \text{SS}_p^{\text{pr}}(R) \\ [\mathfrak{a}], (E, \iota) &\mapsto (E_{\mathfrak{a}}, (\varphi_{\mathfrak{a}})_*(\iota)) \end{aligned}$$

is free with at most two orbits, yielding a free transitive action when restricted to one orbit (see [35, Thm. 10.3.5], [6, § 2.3.4], [17, Thm. 3.1], [16, Thm. 5.4.1], [41, Prop. 3.3 and Thm. 3.4]).

The vectorisation problem for this action, i.e. finding $[\mathfrak{a}]$ given $(E, \iota), (E', \iota')$, is believed quantum-resistant. The most prominent example is CSIDH [14], which uses $R = \mathbb{Z}[\sqrt{-p}]$ and maps $\sqrt{-p}$ to Frobenius. Kuperberg described a subexponential quantum attack [34] on the vectorisation problem, resulting in estimates of 2000-bit primes for NIST-1 security [8, 11, 44].

2.2 Polarised isogenies between abelian varieties

Every abelian variety A has a dual $\widehat{A}$, and isogenies $\varphi: A \rightarrow B$ have duals $\widehat{\varphi}: \widehat{B} \rightarrow \widehat{A}$. A *principal polarisation* is an isomorphism $\lambda_A: A \rightarrow \widehat{A}$ satisfying $\lambda_A \cong \lambda_A$. We call the pair (A, λ_A) a *principally polarised abelian variety* (PPAV); elliptic curves are 1-dimensional PPAVs with canonical polarisations.

Definition 1 (Polarised Isogenies). *For PPAVs $(A, \lambda_A), (B, \lambda_B)$, an isogeny $\varphi: A \rightarrow B$ is (λ_A, λ_B) -polarised of degree d if $\widehat{\varphi}\lambda_B\varphi = [d]\lambda_A$. Such an isogeny is also called a d -isogeny. Its polarised dual is $\widetilde{\varphi} = \lambda_A^{-1}\widehat{\varphi}\lambda_B$, satisfying $\widetilde{\varphi}\varphi = [d]$.*

Lemma 1. *Let $\varphi: (A, \lambda_A) \rightarrow (B, \lambda_B)$ be a d -isogeny between g -dimensional PPAVs over k with $\text{char}(k) \nmid d$. Then:*

- $\ker(\varphi) \subseteq A[d]$ is isotropic for the polarized Weil pairing $e_d^{\lambda_A}$, and has order d^g .
- Conversely, any $\varphi: A \rightarrow B$ with such a kernel induces a unique principal polarisation λ_B making φ a d -isogeny.

Proof. The first claim follows from Weil pairing properties. For the converse, apply [37, Proposition 16.8] to construct λ_B , which must be principal by degree considerations. $\square$

We recall Kani's lemma, which gives a simple way of embedding isogenies in higher-degree isogenies.

Definition 2 (Isogeny diamond). *An (a, b) -isogeny diamond for $a, b \in \mathbb{N}^*$ is a commutative diagram of isogenies between principally polarized abelian varieties A, B, A', B'*

$$\begin{array}{ccc} A' & \xrightarrow{\varphi'} & B' \\ \psi \uparrow & & \uparrow \psi' \\ A & \xrightarrow{\varphi} & B \end{array}$$

where φ, φ' are a -isogenies and ψ, ψ' are b -isogenies.

Kani's lemma states that isogeny diamonds induces polarized isogenies between products of varieties, and conversely, that every polarized isogeny between products of varieties comes from an isogeny diamond. We mainly need the first direction:

Lemma 2 (Kani). *Given an isogeny diamond as in Definition 2, assume that $M := a + b$ is coprime to $\text{char}(k)$, then the isogeny $F : A \times B' \rightarrow B \times A'$ given in matrix form*

$$F := \begin{pmatrix} \varphi & \tilde{\psi}' \\ -\psi & \tilde{\varphi}' \end{pmatrix}$$

is an M -isogeny for the product polarisations. Furthermore, if $\gcd(a, b) = 1$, the kernel of F is

$$\ker(F) = \{(\tilde{\varphi}(x), \psi'(x)) \mid x \in B[M]\} = \{([a]x, \psi' \circ \varphi(x)) \mid x \in A[M]\}.$$

Kani's lemma shows that for $\gcd(a, b) = 1$ it suffices to know $\psi' \circ \varphi$ on $A[M]$ to determine the kernel of F . Furthermore, when M is smooth (in practice a power of 2), we can compute F in polynomial time and thus we can evaluate φ everywhere since $F(x, 0) = (\varphi(x), -\psi(x))$.

2.3 Clapoti, KLaPoTi, and Qlapoti

Given an elliptic curve $E/\mathbb{F}_q$ with a primitive R -orientation, and an invertible ideal $\mathfrak{a} \subset R$, Clapoti [42] gives a general strategy to compute the class group action $E \rightarrow E_{\mathfrak{a}} := E/E[\mathfrak{a}]$, by applying Kani's lemma to the following isogeny diamond:

$$\begin{array}{ccc} E_{\bar{\mathfrak{a}}} & \xrightarrow{\hat{\phi}_{\bar{\mathfrak{b}}}} & E \\ \phi_{\bar{\mathfrak{c}}} \uparrow & & \uparrow \hat{\phi}_{\mathfrak{c}} \\ E & \xrightarrow{\phi_{\mathfrak{b}}} & E_{\mathfrak{a}} \end{array}$$

where $\mathfrak{b}, \mathfrak{c}$ are ideals equivalent to $\mathfrak{a}$ whose norms sum to a smooth integer M (in practice a power of 2). Note that since $\phi_{\mathfrak{a}}$ is only determined up to post-composition with an isomorphism, the above diagram assumes consistent choices so that it becomes commutative. We recall [42, Proposition 2.1]:

Proposition 1. *Let $\mathfrak{b}, \mathfrak{c}$ be ideals equivalent to $\mathfrak{a}$, with $\gcd(n(\mathfrak{b}), n(\mathfrak{c})) = 1$ and $n(\mathfrak{b}) + n(\mathfrak{c}) = M$. Then, the kernel of the M -isogeny*

$$\Phi = \begin{pmatrix} \phi_{\mathfrak{b}} & \phi_{\mathfrak{c}} \\ -\phi_{\bar{\mathfrak{c}}} & \phi_{\bar{\mathfrak{b}}} \end{pmatrix} : E \times E \rightarrow E_{\mathfrak{a}} \times E_{\bar{\mathfrak{a}}}$$

is given by

$$\{([n(\mathfrak{b})]P, \gamma(P) \mid P \in E[M]\},$$

where $\gamma = \widehat{\phi}_{\mathfrak{c}} \circ \phi_{\mathfrak{b}}$ is (the image of) a generator of the principal ideal $\bar{\mathfrak{c}}\mathfrak{b}$.

Although originally formulated for the quadratic case, the above proposition applies readily to quaternion ideals when considering the full ring $\text{End}(E)$ of a supersingular curve, as for instance done in **SQIsign** [5] (although the isogenies on the left and on top no longer correspond to the conjugate ideals). However, given a random ideal $\mathfrak{a}$, it is not clear how to find the ideals $\mathfrak{b}, \mathfrak{c}$ of coprime norms that sum to a smooth integer M (e.g. a power of 2), which led to a necessity of using extra, auxiliary isogenies of large degree.

In KLaPoTi [43], it was shown that the ideals can be found with the KLPT algorithm, thus avoiding the need for the auxiliary isogenies. However, this requires $M \gtrsim \Delta_R^3$ which makes the practicality of the construction rather limited.

However, when looking at quaternion ideals, **Qlapoti** recently showed how to find the ideals $\mathfrak{b}, \mathfrak{c}$ directly, solving the norm equation whenever $M \gtrsim \Delta_R$. This was then applied to **SQIsign**, resulting in significant speed-ups, and a less error-prone algorithm [12].

2.4 Computing higher dimensional isogenies with theta coordinates

As in **Pegasis** [24], we use level 2 theta coordinates due to Mumford [39] to compute a 4-dimensional isogeny obtained from Kani's lemma (see Section 3.1). The approach follows from [21] and we refer to [22, Chapter 6] for a complete exposition.

Symplectic isomorphisms and basis. Let (A, λ_A) be a PPAV of dimension g defined over an algebraically closed field k . Let $n \in \mathbb{N}^*$ coprime with k . Then we know that $A[n] \simeq (\mathbb{Z}/n\mathbb{Z})^{2g}$. We denote by $\widehat{(\mathbb{Z}/n\mathbb{Z})^g}$ the group of characters on $(\mathbb{Z}/n\mathbb{Z})^g$ and define a pairing on $(\mathbb{Z}/n\mathbb{Z})^g \times \widehat{(\mathbb{Z}/n\mathbb{Z})^g}$ by:

$$e_n((i_1, \chi_1), (i_2, \chi_2)) := \chi_2(i_1) \chi_1(i_2)^{-1}.$$

A *symplectic isomorphism* of level n is a group isomorphism $\phi : (\mathbb{Z}/n\mathbb{Z})^g \times \widehat{(\mathbb{Z}/n\mathbb{Z})^g} \xrightarrow{\sim} A[n]$ that respects the n -th Weil pairing $e^{[n]\lambda_A}$ and the natural pairing e_n of $(\mathbb{Z}/n\mathbb{Z})^g \times \widehat{(\mathbb{Z}/n\mathbb{Z})^g}$, as follows:

$$\forall x, y \in (\mathbb{Z}/n\mathbb{Z})^g \times \widehat{(\mathbb{Z}/n\mathbb{Z})^g}, \quad e^{[n]\lambda_A}(\phi(x), \phi(y)) = e_n(x, y).$$

A symplectic isomorphism of level n is always determined by a ζ -symplectic basis of $A[n]$ for some primitive n -th root of unity ζ . Such a basis $(S_1, \dots, S_g, T_1, \dots, T_g)$ satisfies for all $l, m \in \llbracket 1 ; g \rrbracket$,

$$e^{[n]\lambda_A}(S_l, S_m) = e^{[n]\lambda_A}(T_l, T_m) = 1 \quad \text{and} \quad e^{[n]\lambda_A}(S_l, T_m) = \zeta^{\delta_{l,m}}.$$

We shall drop the mention of ζ when it is not specified.

Theta structures. For the sake of clarity, we define theta structures as in [30, Definition 4], which requires much less algebraic geometry background than the original definition from Mumford [39].

Definition 3. A theta structure of level n on a PPAV (A, λ_A) is a map to the projective space $\Theta_n : A \rightarrow \mathbb{P}^{n^g-1}, x \mapsto (\theta_i(x))_{i \in (\mathbb{Z}/n\mathbb{Z})^g}$ along with a symplectic isomorphism $\overline{\Theta}_n : (\mathbb{Z}/n\mathbb{Z})^g \times (\mathbb{Z}/n\mathbb{Z})^g \xrightarrow{\sim} A[n]$ such that for all $x \in A[n]$, $(i, \chi) \in (\mathbb{Z}/n\mathbb{Z})^g \times (\mathbb{Z}/n\mathbb{Z})^g$ and $j \in (\mathbb{Z}/n\mathbb{Z})^g$:

$$\theta_i(x + \overline{\Theta}_n(i, \chi)) = \chi(i + j)^{-1} \theta_{i+j}(x). \quad (3)$$

The θ_i are called the theta coordinates associated to the theta structure Θ_n , or simply Θ_n -coordinates.

A theta structure defines an embedding $A \hookrightarrow \mathbb{P}^{n^g-1}$ when $n \geq 3$ [40, p. 163] and induces an embedding of the Kummer variety $A/\pm \hookrightarrow \mathbb{P}_k^{2^g-1}$ when $n = 2$ and (A, λ_A) is not a polarised product [10, Theorem 4.8.1]. Either way theta coordinates define a system of coordinates on the abelian variety or its Kummer variety. In practice, we work in level $n = 2$ so theta coordinates represent points on the Kummer variety *i.e.* points up to sign.

We define the *theta null point* as $(\theta_i(0_A))_i$ and call its theta coordinates the *theta constants*. When $4|n$, the theta null point fully determines the PPAV together with its theta structure (A, λ_A, Θ_n) [39, Corollary p. 340] so it can be used to represent this data on a computer. Even when $n = 2$, the theta null point is sufficient in most cases to represent A since the theta null point can be used for common arithmetic operations, including point duplication $x \mapsto 2x$ and differential addition $x, y, x - y \mapsto x + y$ (see [45, Algorithm 4.4.10]).

When it is *symmetric* in the sense of [39, Definition p. 317], a theta structure Θ_n of level n on (A, λ_A) is fully determined by a symplectic basis of $A[2n]$ (whose double induces $\overline{\Theta}_n$) [39, Remark 3, p. 319]. For this convenient way to represent theta structures, we shall always assume that they are symmetric.

Change of theta coordinates. We shall see that we need a change of theta structure and to compute the associated change of theta coordinates for isogeny computations. Let Θ_n and Θ'_n be two (symmetric) theta structures of level n on a PPAV (A, λ_A) . We know that Θ_n and Θ'_n are both induced by ζ -symplectic basis of $A[2n]$ (where $\zeta \in k^*$ is a primitive $2n$ -th root of unity) that we shall denote by $\mathcal{B}$ and $\mathcal{B}'$ respectively. From the symplectic change

of basis matrix $M \in \mathrm{Sp}_{2g}(\mathbb{Z}/2n\mathbb{Z})$ from $\mathcal{B}$ to $\mathcal{B}'$ and ζ , we can compute a matrix $N(M, \zeta) \in M_{n^g}(\mathbb{Z}[\zeta])$ to obtain the Θ'_n -coordinates $(\theta'_i)_i$ from the Θ_n -coordinates $(\theta_i)_i$ as follows: $(\theta'_i)_i = N(M, \zeta) \cdot (\theta_i)_i$. The exact formula may be found in [22, Theorem 6.2.10].

Applying the following symplectic change of basis that swaps the two sides of the basis

$$M := \begin{pmatrix} 0 & -I_g \\ I_g & 0 \end{pmatrix},$$

corresponds to a *Hadamard transform* on theta coordinates that yields the *dual theta coordinates* of $(\theta_i)_i$ given by:

$$\forall i \in (\mathbb{Z}/n\mathbb{Z})^g, \quad U_i := \sum_{j \in (\mathbb{Z}/n\mathbb{Z})^g} \zeta^{2\langle i|j \rangle} \theta_j,$$

where $\langle \cdot | \cdot \rangle$ is the usual scalar product.

Theta structures on a Montgomery elliptic curve. In qt-Pegasis, as in Pegasis, we compute isogenies obtained by Kani's lemma and defined on products of elliptic curves. So we need to compute theta structures on such products. We start by explaining how to obtain a theta structure on a Montgomery elliptic curve. Let E be such a curve. Then from a basis (P, Q) of $E[4]$ with Montgomery $(x : z)$ -coordinates $x(Q) = -z(Q)$, we obtain a level 2 theta structure Θ_E with theta null point $(a : b) = (x(P) + z(P) : x(P) - z(P))$, given by $(\theta_0^E : \theta_1^E) = (a(x - z) : b(x + z))$ [46, Chapter 7, Appendix A]. This means in particular that the theta structure is defined over the field of definition of the $(x : z)$ -coordinates of (P, Q) .

Product theta structures. Let $\Theta_{A_1}, \dots, \Theta_{A_r}$ be level n theta structures respectively defined on PPAVs $(A_1, \lambda_1), \dots, (A_r, \lambda_r)$ of dimensions $g_1, \dots, g_r$. The *product theta structure* $\Theta_{A_1 \times \dots \times A_r} = \Theta_{A_1} \times \dots \times \Theta_{A_r}$ is the theta structure with theta coordinates given by:

$$\theta_{i_1, \dots, i_r}^{A_1 \times \dots \times A_r}(x_1, \dots, x_r) = \prod_{m=1}^r \theta_{i_m}^{A_m}(x_m),$$

for all $(x_1, \dots, x_r) \in A_1 \times \dots \times A_r$, $i_1 \in (\mathbb{Z}/n\mathbb{Z})^{g_1}, \dots, i_r \in (\mathbb{Z}/n\mathbb{Z})^{g_r}$. If Θ_{A_m} is determined by a ζ -symplectic basis $\mathcal{B}_m := (S_1^{(m)}, \dots, S_{g_m}^{(m)}, T_1^{(m)}, \dots, T_{g_m}^{(m)})$ of $A_m[n]$ for all $m \in \llbracket 1 ; r \rrbracket$, then their product is determined by the *product ζ -symplectic basis*:

$$\begin{aligned} \mathcal{B}_1 \times \dots \times \mathcal{B}_r := & ((S_1^{(1)}, \dots, 0), \dots, (S_{g_1}^{(1)}, \dots, 0), \dots, \\ & (0, \dots, S_1^{(r)}), \dots, (0, \dots, S_{g_r}^{(r)}), (T_1^{(1)}, \dots, 0), \dots, (T_{g_1}^{(1)}, \dots, 0), \dots, \\ & (0, \dots, T_1^{(r)}), \dots, (0, \dots, T_{g_r}^{(r)})). \end{aligned}$$

Unsurprisingly, this product theta structure is the one we naturally obtain on elliptic products.

How to compute a 2^e -isogeny. Let $F : (A, \lambda_A) \rightarrow (B, \lambda_B)$ be a 2^e -isogeny between PPAVs of dimension g . Assume that we are given generators of an isotropic subgroup $K \subset A[2^{e+2}]$ such that $\ker(F) = [4]K$ and a level 2 theta structure Θ_A on A . For instance, Θ_A may be a product theta structure when F is obtained from Kani's lemma. Then, following [21] or [22, Chapter 6] we can compute F as follows:

- **Step A:** From generators of K , we can obtain a symplectic basis $\mathcal{B} := (S_1, \dots, S_g, T_1, \dots, T_g)$ of $A[2^{e+2}]$ that is *adapted to F* i.e. such that $\ker(F) = \langle [4]T_1, \dots, [4]T_g \rangle$. When F is derived from Kani's lemma, $\mathcal{B}$ can be obtained from [22, Lemmas 6.4.1 and 6.4.3].
Let $\mathcal{B}_0$ be a symplectic basis of $A[4]$ inducing Θ_A and let Θ'_A be the level 2 theta structure induced by $[2^e]\mathcal{B}$. We can then compute the symplectic change of basis matrix from $\mathcal{B}_0$ to $[2^e]\mathcal{B}$ and apply [22, Theorem 6.2.10] to express Θ'_A -coordinates from Θ_A -coordinates.
- **Step B:** From $T_1, \dots, T_g$ expressed in these Θ'_A -coordinates, we can compute F decomposed as a chain of 2-isogenies $F := f_e \circ \dots \circ f_1$. For all $m \in \llbracket 1 ; e \rrbracket$, f_m can be computed from $[2^{e-m}]f_{m-1} \circ \dots \circ f_1(T_1, \dots, T_g)$ [22, Lemmas 6.3.1 and 6.3.4]. To compute f_m , it suffices to compute the dual theta null point of its codomain, using [22, Algorithm 6.5]. This 2-isogeny can then be evaluated with [22, Algorithm 6.1]. Quasi-linear strategies can be applied to minimize the number of point duplications and evaluations required [22, § 6.3.3].
Note that the first isogenies of the chain may be gluing isogenies i.e. isogenies defined over a product of PPAVs or have other kind of singularities (see Section 4.2 in the case of qt-Pegasis). In these cases, specific codomain computation and evaluation algorithms have to be used (see [22, § 6.1.3] or [23]).
- **Step C:** When F is obtained from Kani's lemma, (B, λ_B) is generally a product of PPAVs and we need to recover a level 2 product theta structure Θ_B on (B, λ_B) to decompose it and express F component-wise. The symplectic basis $\mathcal{B}$ adapted to F together with F naturally induce the level 2 theta structure Θ'_B that we obtain at the end of the chain computation. By [22, Theorem 6.1.1], Θ'_B is induced by the symplectic basis $F_*(\mathcal{B}) := ([2^e]F(S_1), \dots, [2^e]F(S_g), F(T_1), \dots, F(T_g))$ of $B[4]$. Knowing a change of basis matrix from $F_*(\mathcal{B})$ to a symplectic basis $\mathcal{C}_0$ inducing Θ_B , we can use [22, Theorem 6.2.10] to compute Θ_B -coordinates from Θ'_B -coordinates, as desired. This change of basis matrix can be computed with [22, Lemmas 6.4.1 and 6.4.3] when F is derived from Kani's lemma.

3 The qt-Pegasis algorithm

We now describe our new algorithm. Throughout this section, let $R = \mathbb{Z}[\omega]$ be an imaginary quadratic order, let E be a supersingular elliptic curve, and let $\tau \in \text{End}(E)$ be an endomorphism such that

$$\iota : R \hookrightarrow \text{End}(E)$$

$$\iota(a + b\omega) = [a] + [b]\tau$$

is a primitive R -orientation on E . We assume that E is maximal and defined over $\mathbb{F}_{p^2}$ (or even $\mathbb{F}_p$) with $p = c2^f - 1$ and c small, so that $E[2^f] \subseteq E(\mathbb{F}_{p^2})$.

3.1 A nested Kani-diamond

Our starting point comes from the observation that given four ideals $\mathfrak{b}_1, \mathfrak{b}_2, \mathfrak{c}_1, \mathfrak{c}_2$ all equivalent to $\mathfrak{a}$, we can apply Kani's lemma (Lemma 2) twice to construct an isogeny, where we can recover the kernel in a similar way as in Clapoti (Proposition 1). As we will see, finding $\mathfrak{b}_1, \mathfrak{b}_2, \mathfrak{c}_1, \mathfrak{c}_2$ can naturally be done with a simplified and optimized version of Qlapoti.

Proposition 2. *Let $\mathfrak{b}_1, \mathfrak{b}_2, \mathfrak{c}_1, \mathfrak{c}_2$ be ideals equivalent to $\mathfrak{a} \subseteq R$, let $N_1 = n(\mathfrak{b}_1) + n(\mathfrak{c}_1)$ and $N_2 = n(\mathfrak{b}_2) + n(\mathfrak{c}_2)$ and assume that $\gcd(N_1, N_2) = 1$ and let $M = N_1 + N_2$. Then, the kernel of the M -isogeny induced by the Kani diamond*

$$\begin{array}{ccc} A & \xrightarrow{\psi_2} & E \times E \\ \psi_1 \uparrow & & \uparrow \Phi_2 \\ E \times E & \xrightarrow{\Phi_1} & E_{\mathfrak{a}} \times E_{\bar{\mathfrak{a}}} \end{array}$$

where

$$\Phi_1 := \begin{pmatrix} \phi_{\mathfrak{b}_1} & \phi_{\mathfrak{c}_1} \\ -\phi_{\bar{\mathfrak{c}_1}} & \phi_{\bar{\mathfrak{b}_1}} \end{pmatrix}, \quad \tilde{\Phi}_2 := \begin{pmatrix} \phi_{\mathfrak{b}_2} & \phi_{\mathfrak{c}_2} \\ -\phi_{\bar{\mathfrak{c}_2}} & \phi_{\bar{\mathfrak{b}_2}} \end{pmatrix},$$

is given by

$$\{([N_1]P, [N_1]Q, \gamma_1(P) + \gamma_2(Q), \bar{\gamma}_1(Q) - \bar{\gamma}_2(P)) \mid (P, Q) \in (E \times E)[M]\}$$

where $\gamma_1 := \phi_{\bar{\mathfrak{b}_2}} \circ \phi_{\mathfrak{b}_1} + \phi_{\mathfrak{c}_2} \circ \phi_{\bar{\mathfrak{c}_1}}$ and $\gamma_2 := \phi_{\bar{\mathfrak{b}_2}} \circ \phi_{\mathfrak{c}_1} - \phi_{\mathfrak{c}_2} \circ \phi_{\bar{\mathfrak{b}_1}}$ are in $R \subseteq \text{End}(E)$.

Proof. First, note that the isogenies Φ_1, Φ_2 are principally polarized by Kani's lemma. In fact, they are exactly the isogenies from Proposition 1, and it follows that they have degree N_1 and N_2 respectively.

Then applying Kani's lemma again gives the 4-dimensional isogeny

$$F := \begin{pmatrix} \Phi_1 & \tilde{\Phi}_2 \\ -\Psi_1 & \tilde{\Psi}_2 \end{pmatrix} : (E \times E) \times (E \times E) \rightarrow (E_{\mathfrak{a}} \times E_{\bar{\mathfrak{a}}}) \times A,$$

with kernel

$$\ker F = \{([N_1](P, Q), \Phi_2 \circ \Phi_1((P, Q))) \mid (P, Q) \in (E \times E)[M]\},$$

which gives the final description of the kernel by explicit computation. The fact that γ_1 and γ_2 are in R is clear from the fact that $\phi_{\mathfrak{b}_i}$ and $\phi_{\mathfrak{c}_i}$ are isogenies coming from R -ideals all equivalent to the ideal $\mathfrak{a}$. $\square$

Remark 1. In practice we will take $M = 2^e$, and since we require $\gcd(N_1, N_2) = 1$ and $N_1 + N_2 = 2^e$, this implies that both N_1, N_2 have to be odd. Although it is possible to deal with even N_i (note that the 2-valuation of both automatically is the same) by using 2-dimensional 2-isogenies, it turns out to be more efficient to simply look for ideals resulting in odd N_i .

Thus, we are left with the task of finding the ideals $\mathfrak{b}_1, \mathfrak{b}_2, \mathfrak{c}_1, \mathfrak{c}_2$. We now show how this problem is solved by the algorithm introduced in **Qlapoti**.

3.2 Solving the norm equation

Given an invertible ideal $\mathfrak{a} \subset R$, we start by writing $\mathfrak{a} = (N, \alpha)$, with $N = \mathfrak{n}(\mathfrak{a})$, and $\mathfrak{n}(\alpha)$ as small as possible by lattice reduction. We are free to replace $\mathfrak{a}$ by an equivalent ideal, thus we assume that N is the norm of the smallest ideal in the class $[\mathfrak{a}]$. Define $\delta = N/\sqrt{-\Delta_R}$, then by Minkowski we know that $\delta < 2/\pi$ and also define $\epsilon = 2^e / -\Delta_R$. Note that since $\mathfrak{a}$ is the smallest norm ideal in its class, it must be primitive, and thus $\gcd(N, \text{tr}(\alpha)) = 1$. The ideals $\mathfrak{b}_i$ and $\mathfrak{c}_i$ are equivalent to $\mathfrak{a}$ and hence of the form $\mathfrak{a}\overline{\alpha_i}/N$ for $\alpha_i \in \mathfrak{a}$. Thus our goal is to find four elements $\beta_1, \beta_2, \delta_1, \delta_2$ all in $\mathfrak{a}$ solving the norm equation

$$\mathfrak{n}(\beta_1) + \mathfrak{n}(\beta_2) + \mathfrak{n}(\delta_1) + \mathfrak{n}(\delta_2) = 2^e N. \quad (4)$$

We now briefly review why the above norm equation can already be solved by **Qlapoti**. Notice that given an ideal $\mathfrak{a} \subset R$ we can construct the quaternion order $R + \mathbf{i}R$, defined by the multiplicative relations $\mathbf{i}^2 = -1$ and $\mathbf{i}\omega = \overline{\omega}\mathbf{i}$. This allows us to embed $\mathfrak{a}$ into the quaternion ideal $\mathfrak{a} + \mathbf{i}\mathfrak{a} \subseteq R + \mathbf{i}R$, and apply **Qlapoti**. Now **Qlapoti** outputs two quaternions $\gamma_1, \gamma_2 \in \mathfrak{a} + \mathbf{i}\mathfrak{a}$ such that $\text{nrd}(\gamma_1) + \text{nrd}(\gamma_2) = 2^e N$. However, we can rewrite $\gamma_i = \beta_i + \mathbf{i}\delta_i$, and since every element of $\mathbf{i}R$ is trace free, we have

$$\text{nrd}(\gamma_i) = \text{nrd}(\beta_i) + \text{nrd}(\delta_i) + \text{trd}(\beta_i \mathbf{i} \overline{\delta_i}) = \text{nrd}(\beta_i) + \text{nrd}(\delta_i),$$

and thus, we end up with four elements $\beta_1, \beta_2, \delta_1, \delta_2 \in \mathfrak{a}$, satisfying Equation (4).

Remark 2. This is essentially exactly the same observation that forms the basis of KLaPoTi. There, the goal is to find two equivalent quadratic ideals $\mathfrak{b}, \mathfrak{c}$ satisfying $\mathfrak{n}(\mathfrak{b}) + \mathfrak{n}(\mathfrak{c}) = 2^e$, which is solved by finding a single quaternion ideal $I \sim \mathfrak{a} + \mathbf{i}\mathfrak{a}$ satisfying $\text{nrd}(I) = 2^e$ using the KLPT algorithm (thus, they require $2^e > \Delta_R^3$). Our setup is almost identical, except we double the number of both quadratic and quaternion ideals. The main difference is that we do not rely on KLPT, but apply techniques from **Qlapoti**.

Instead of simply applying **Qlapoti**, we exploit the fact that the quaternion ideals obtained above are not random quaternion ideals, but very specific ones, coming from the optimal embedding of R into $R + \mathbf{i}R$. For instance, when $\Delta_R = p \equiv 3 \pmod{4}$, R corresponds to the CSURF-orientation, and $R + \mathbf{i}R$ can be identified with $\text{End}(E_0)$, where $j(E_0) = 1728$, and the ideals obtained this way

are precisely the ideals corresponding to horizontal isogenies (thus, in particular, defined over $\mathbb{F}_p$). We now discuss how to leverage this specific “shape” of ideals to create an even simpler variant of **Qlapoti**.

To find $\mathbf{c}_1, \mathbf{c}_2$, we look for elements $\delta_i \in \mathfrak{a}$ of the form $c_i N + d_i \alpha$, while for $\mathbf{b}_1, \mathbf{b}_2$ we simply look for elements of the form $b_i N$. Let us denote $\mathfrak{n}(\alpha) = rN$. The norm equation then reads

$$N(b_1^2 + b_2^2 + c_1^2 + c_2^2) + r(d_1^2 + d_2^2) + \text{tr}(\alpha)(c_1 d_1 + c_2 d_2) = 2^e. \quad (5)$$

Although similar, it turns out that this particular norm equation can be solved by an even more efficient procedure than the norm equation from **Qlapoti**. We apply essentially the same 2-step strategy as **Qlapoti**: in the first step, we reduce the above equation modulo N and find small solutions for (c_1, c_2, d_1, d_2) (in particular, each will be smaller than $\sqrt{N}$) in general using lattice reduction. In the second step, we substitute these solutions in the above equation, and solve the remaining sums-of-squares problem for (b_1, b_2) using Cornacchia’s algorithm.

Setting up the lattice. We start by setting $d_1 = 1$ and $d_2 = k$, where we will vary k to get a small enough solution for c_1, c_2 when solving the above equation modulo N . Ignoring cross terms, Equation (5) will clearly only admit solutions for $(1 + k^2)r < 2^e$, so we should take k smaller than $\sqrt{2^e/r}$. Define $k_{\max} = \min\{\sqrt{2^e/r}, \sqrt{N}\}$, then we will choose $k < k_{\max}$. For small discriminants, the latter term becomes the minimum and will guarantee a certain vector to be the shortest vector.

As will become clear below, we will choose k close to this upper bound (which will increase the balancedness of a certain lattice), and then reduce the above equation modulo N and obtain

$$c_1 + c_2 k = (2^e - (1 + k^2)r) \text{tr}(\alpha)^{-1} \bmod N,$$

where, as pointed out earlier, $\text{tr}(\alpha)$ is indeed invertible modulo N . Define the right hand side $v = (2^e - (1 + k^2)r) \text{tr}(\alpha)^{-1} \bmod N$, then it is clear that $(v, 0)$ is a solution to the above equation, but not very short. To make it shorter, we subtract a close vector in the lattice of solutions to the homogeneous equation $c_1 + c_2 k = 0 \bmod N$, which is generated by the rows of the matrix

$$\begin{pmatrix} k & -1 \\ N & 0 \end{pmatrix},$$

and has volume N . Furthermore, since $k < k_{\max} < \sqrt{N}$, the first vector is expected to be the shortest vector in the lattice, and a second independent short vector can be obtained by size reducing the second one with respect to the first one.

A second short basis vector can be found by determining the integer $x > 0$ that minimizes the norm of the vector $(N - kx, x)$. The norm will be minimal when $N - kx = x$, or $x = N/(k + 1)$, but since x needs to be an integer we

get $x = \lfloor N/(k+1) \rfloor$. The 2-norm of the above vector then becomes $\sim \sqrt{2}N/k$, which makes it $\sqrt{2}N/k^2$ times longer than the smallest vector. To make the lattice balanced, i.e. having roughly equal lattice minima, it is therefore best to take k as large as possible, whilst still satisfying $k < k_{\max}$.

To compute a close vector to $(v, 0)$ we could use Babai rounding given the 2 short vectors above, which requires computing the inverse of a 2×2 matrix, a rounding and a matrix multiply. However, our observation is that the vector $(v, 0)$ will be close to orthogonal to the second reduced basis vector, and thus the CVP solution we are looking for is close to being just a multiple of the first basis vector $[k, -1]$ (see Figure 1). We now discuss this idea in more detail.

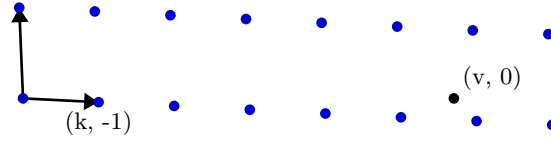


Fig. 1. By construction, the closest vector to $(v, 0)$ is typically a multiple of $[k, -1]$.

Looking for good CVP instances. A much simpler, but a priori, cruder approach than Babai rounding is to compute a close vector by computing the closest multiple of the shortest vector $(k, -1)$ to $(v, 0)$. In particular, if we let $y = \lfloor v/k \rfloor$, then $(yk, -y)$ is close to $(v, 0)$, and we can estimate the norm of the difference

$$\|(v - yk, y)\|_2 \approx \sqrt{k^2/4 + (v/k)^2}.$$

The above short solution $(s, t) = (v - yk, y)$ for (c_1, c_2) will potentially result in a solution (for (b_1, b_2)) as long as its squared norm is smaller than $2^e/N$, i.e. it suffices that v satisfies

$$v < k\sqrt{2^e/N - k^2/4}. \quad (6)$$

Our strategy therefore is very simple: we start by setting $k_0 = k_{\max}$, and compute the first $v = v_0$ as

$$v_0 \equiv (2^e - r(1 + k_0^2)) \text{tr}(\alpha)^{-1} \pmod{N}.$$

Now, for each step i , if v_i satisfies the bound $v_i < k_{\max}\sqrt{2^e/N - k_{\max}^2/4}$, we know that we will get a sufficiently short solution. Otherwise, we update v_{i+1} by decrementing $k_{i+1} = k_i - 1$. Thus the new value v_{i+1} becomes

$$\begin{aligned} v_{i+1} &\equiv (2^e - r(1 + (k_i - 1)^2)) \text{tr}(\alpha)^{-1} \pmod{N} \\ &\equiv (2^e - r(1 + k_i^2) + r(2k_i - 1)) \text{tr}(\alpha)^{-1} \pmod{N} \\ &\equiv v_i + r(2k_i - 1) \text{tr}(\alpha)^{-1} \pmod{N} \end{aligned}$$

Thus, for each loop, we can update v_i simply using 2 additions modulo N (one addition to update the offset $r(2k_i - 1) \text{tr}(\alpha)^{-1}$ and one addition to update v_i), and whenever $v_i < k_{\max} \sqrt{2^e/N - k_{\max}^2/4}$, we compute the above closest vector to obtain a short solution for (c_1, c_2) satisfying $c_1 + c_2 k = v_i \pmod{N}$.

The number of iterations required to find a good k_i mainly depends on the relative size of the discriminant versus 2^e . The stopping condition is given by Equation (6), so if we consider the v_i to be random elements modulo N , the expected number of iterations is given by the inverse of

$$\frac{k_{\max}}{N} \sqrt{\frac{2^e}{N} - \frac{k_{\max}^2}{4}},$$

since all $k_i \simeq k_{\max}$. Recall that we defined $\delta = N/\sqrt{-\Delta_R} < 2/\pi$ and $\epsilon = 2^e / -\Delta_R$. Further, we defined $\mathfrak{a}$ to be the ideal with the smallest norm in $[\mathfrak{a}]$, and α to be a generator $\mathfrak{a} = (N, \alpha)$ with $n(\alpha) = Nr$ as small as possible. Minkowski's second theorem implies that $n(\xi_1)n(\xi_2) < -(4/\pi^2)\Delta_R N^2$ where ξ_1 is an element of shortest norm and ξ_2 a $\mathbb{Q}$ -linearly independent element of smallest norm. Since in our case $n(\xi_1) = N^2$ and $n(\xi_2) = Nr$, we thus have $r < -(4/\pi^2)\Delta_R/N$. For the case $k_{\max} = \sqrt{2^e/r} > \sqrt{\epsilon' N}$ with $\epsilon' = \epsilon\pi^2/4$ we therefore expect to require

$$\begin{aligned} & \left(\frac{k_{\max}}{N} \sqrt{\frac{2^e}{N} - \frac{k_{\max}^2}{4}} \right)^{-1} \\ & < \left(\sqrt{\frac{\epsilon'}{N}} \sqrt{\frac{2^e}{N} - \frac{\epsilon' N}{4}} \right)^{-1} = \left(\sqrt{\frac{4\epsilon\epsilon'(-\Delta_R) - \epsilon'^2 N^2}{4N^2}} \right)^{-1} \\ & = \sqrt{\frac{4N^2}{4\epsilon\epsilon' \frac{N^2}{\delta^2} - \epsilon'^2 N^2}} = \sqrt{\frac{4\delta^2}{\epsilon'^2(16/\pi - \delta^2)}} = \epsilon^{-1} \frac{8\delta}{\pi^2 \sqrt{16/\pi^2 - \delta^2}} \end{aligned}$$

iterations. Since $\delta < 2/\pi$, we see that this is upper bounded by $\epsilon^{-1} \cdot 1.470$. In particular, for $\epsilon > 1$, we expect to need a single iteration on average. In the CSURF case, we have $p = c2^{e+3} - 1$ and $\Delta_R = -p$, so in this case $\epsilon^{-1} \simeq 8c$, so the number of iterations is directly proportional to the cofactor c .

Remark 3. We also note that if $k_{\max}$ is smaller than expected, this implies that r is large, which in turn implies that N is small. Since δ is proportional to N , this means that the number of required iterations also decreases proportionally. As in **Qlapoti**, we thus see that the “only” thing that can go wrong, when ϵ is reasonably big, is that r becomes so large that $k_{\max} = 0$. However, this then implies that N is tiny, and $\mathfrak{a}$ can then be translated to its corresponding isogeny directly.

Finalizing the output. We obtain a short pair (c_1, c_2) such that $c_1 + c_2 k = v \pmod{N}$. We then proceed as in the general algorithm by solving the equation

$$b_1^2 + b_2^2 = \frac{2^e - (1 + k^2)r - \text{tr}(\alpha)(c_1 + c_2 k)}{N} - (c_1^2 + c_2^2), \quad (7)$$

for b_1, b_2 by using Cornacchia's algorithm. If the above sum of squares problem is not solvable, we simply go back to searching for new (c_1, c_2) pairs until we hit a valid instance. From here we can directly define

$$\beta_1 = b_1 N, \quad \beta_2 = b_2 N, \quad \delta_1 = c_1 N + \alpha, \quad \delta_2 = c_2 N + k\alpha$$

and

$$\mathbf{b}_1 = \mathbf{a}\bar{\beta}_1/N, \quad \mathbf{b}_2 = \mathbf{a}\bar{\beta}_2/N, \quad \mathbf{c}_1 = \mathbf{a}\bar{\delta}_1/N, \quad \mathbf{c}_2 = \mathbf{a}\bar{\delta}_2/N$$

Note that unlike the original Qlapoti-algorithm, no back-substitution is needed.

We summarize the algorithm in Algorithm 1.

Algorithm 1 SpecialQlapoti

Input: Left R -ideal $\mathbf{a}$, a number $e \in \mathbb{N}$.

Output: Equivalent ideals $\mathbf{b}_1, \mathbf{b}_2, \mathbf{c}_1, \mathbf{c}_2$ with $n(\mathbf{b}_1) + n(\mathbf{b}_2) + n(\mathbf{c}_1) + n(\mathbf{c}_2) = 2^e$.

```

1:  $\mathbf{a} \leftarrow$  Smallest ideal in  $[\mathbf{a}]$ 
2: Write  $\mathbf{a} = (N, \alpha)$ , where  $N$  is the norm of  $\mathbf{a}$ 
3:  $r \leftarrow n(\alpha)/N$ 
4:  $k \leftarrow \min\{\sqrt{2^e/r}, \sqrt{N}\}$ 
5:  $\text{bound} \leftarrow \lfloor k\sqrt{2^e/N - k^2/4} \rfloor$ 
6:  $v \leftarrow (2^e - (1 + k^2)r) \text{tr}(\alpha)^{-1} \pmod{N}$ 
7:  $t_1 \leftarrow r(2k - 1) \text{tr}(\alpha)^{-1} \pmod{N}$ 
8:  $t_2 \leftarrow -2r \text{tr}(\alpha)^{-1} \pmod{N}$ 
9: while true do
10:   if  $v < \text{bound}$  then
11:      $y \leftarrow \lfloor v/k \rfloor, (c_1, c_2) \leftarrow (v - yk, -y)$ 
12:      $z \leftarrow \frac{2^e - (1 + k^2)r - \text{tr}(\alpha)(c_1 + c_2 k)}{N} - (c_1^2 + c_2^2)$ 
13:      $\text{sol} \leftarrow \text{Cornacchia}(z)$ 
14:     if  $\text{sol} \neq \perp$  then
15:        $b_1, b_2 \leftarrow \text{sol}$ 
16:        $\beta_1, \beta_2, \delta_1, \delta_2 \leftarrow b_1 N, b_2 N, c_1 N + \alpha, c_2 N + k\alpha$ 
17:       return  $\mathbf{a}\bar{\beta}_1/N, \mathbf{a}\bar{\beta}_2/N, \mathbf{a}\bar{\delta}_1/N, \mathbf{a}\bar{\delta}_2/N$ 
18:    $k \leftarrow k - 1$ 
19:    $v \leftarrow v + t_1 \pmod{N}$ 
20:    $t_1 \leftarrow t_1 + t_2 \pmod{N}$ 

```

Remark 4. In practice, we will not work with these ideals but instead with principal ideals coming from their products. Generators for these ideals can be computed directly. For instance, a generator for $\mathbf{b}_i \bar{\mathbf{c}}_j$ is $b_i \delta_j$, which is an element of the correct norm in the ideal. Notice that in principle, $-b_i \delta_j$ is also a valid choice; it is however important that the choices we make are consistent with each other, as explained in Remark 6. The only values we will need in the following are (note that these are precisely the ideals occurring in the endomorphisms γ_i

defining the kernel):

$$\begin{aligned}
A_1 + \omega A_2 &= \bar{c}_1 \mathfrak{b}_1 = b_1 \delta_1 \\
B_1 + \omega B_2 &= \bar{b}_2 \mathfrak{b}_1 = b_1 b_2 N \\
C_1 + \omega C_2 &= \mathfrak{c}_2 \bar{\mathfrak{b}}_1 = b_1 \bar{\delta}_2 \\
D_1 + \omega D_2 &= \bar{c}_1 \mathfrak{c}_2 = \delta_1 \bar{\delta}_2 / N \\
E_1 + \omega E_2 &= \bar{\mathfrak{b}}_2 \mathfrak{c}_1 = b_2 \bar{\delta}_1
\end{aligned} \tag{8}$$

Remark 5. We will impose additional conditions on the output of Algorithm 1 to simplify the isogeny computation, e.g. we require that N_i are odd and coprime (see Proposition 2). These conditions are given in Lemma 4 and Lemma 5 and an adaptation of Algorithm 1 to enforce them is discussed in Section 4.3.

4 4-dimensional isogenies

In this section, we explain in more detail how to compute the 4-dimensional 2^e -isogeny $F : E^4 \rightarrow E_{\mathfrak{a}} \times E_{\bar{\mathfrak{a}}} \times A$ from Proposition 2, using the techniques explained in Section 2.4 that we briefly recall here.

- The first step (Step A in Section 2.4) is to compute a symplectic basis $\mathcal{B}$ of $E^4[2^{e+2}]$ which is adapted to F in order to obtain a level 2 theta structure adapted to the first 2-isogeny $f_1 : E^4 \rightarrow \mathcal{A}_1$ on E^4 . This will be explained in Section 4.1.
- The second step (Step B) is the computation of the 2-isogeny chain $f_1, f_2, \dots, f_e$. Recall that the first isogenies of the chain may be gluing isogenies or other isogenies between products of abelian varieties that needs to be computed and evaluated with specific (and more costly) algorithms. We shall impose restricted conditions on our norm equation solutions from Algorithm 1 to ensure that the first isogeny $f_1 : E^4 \rightarrow \mathcal{A}_1$ is a gluing isogeny whose codomain is not a product. Hence, unlike in Pegasis [24] (see Appendix B.1), the following isogenies $f_2, \dots, f_e$ can all be computed with generic algorithms. In addition to algorithmic simplicity, imposing these conditions facilitate a constant time implementation of qt-Pegasis. We introduce these conditions in Section 4.2 and the early rejections in Algorithm 1 needed to satisfy them in Section 4.3. In Section 4.2, we also treat a singularity that may impact the computation of f_2 .
- The third step (Step C) is the recovery of the codomain product theta structure on $E_{\mathfrak{a}} \times E_{\bar{\mathfrak{a}}} \times A$. This is also explained in Section 4.1.

Note that, as in [24], our implementation works in the CSURF context [13] with $p = c2^f - 1$, $R = \mathbb{Z}[\omega]$ and $\omega = \frac{1+\sqrt{-p}}{2}$. While the theta structure choices in Section 4.1 are presented in plain generality before being specialised to CSURF, the study of singularities in Section 4.2 and the related early rejections in Algorithm 1 presented in Section 4.3 are mainly specific to the CSURF context.

4.1 Theta structures

In this section, we compute a symplectic basis $\mathcal{B}$ of $E^4[2^{e+2}]$ inducing a theta structure adapted to F on the domain and the theta structure induced by $F_*(\mathcal{B})$ on the codomain. We also explain how to recover the codomain product theta structure.

In the general context. Let (P, Q) be a basis of $E[2^{e+2}]$ and $\zeta := e_{2^{e+2}}(P, Q)$. Consider the ζ -symplectic basis (x_1, x_2, y_1, y_2) of $E^2[2^{e+2}]$ given by:

$$x_1 := (P, \mathbf{0}), \quad x_2 := (\mathbf{0}, P), \quad y_1 := (Q, \mathbf{0}), \quad y_2 := (\mathbf{0}, Q).$$

Then, by [22, Lemma 6.4.3], we obtain a ζ -symplectic basis $\mathcal{B} := (S_1, \dots, S_4, T_1, \dots, T_4)$ of $E^4[2^{e+2}]$ that is *adapted to F* i.e. such that $\ker(F) = \langle [4]T_1, \dots, [4]T_4 \rangle$. It is given by:

$$\begin{aligned} S_l &:= ([-\alpha]y_l, \mathbf{0}, \mathbf{0}), \quad S_{l+2} := (\mathbf{0}, \mathbf{0}, [\beta]\Phi_2 \circ \Phi_1(x_l)), \\ T_l &:= ([N_1]x_l, \Phi_2 \circ \Phi_1(x_l)), \quad T_{l+g} := ([1 - \alpha 2^e]y_l, [\alpha]\Phi_2 \circ \Phi_1(y_l)), \end{aligned}$$

for $l \in \{1, 2\}$, with $\alpha \equiv N_1^{-1} \bmod 2^{e+2}$ and $\beta \equiv N_2^{-1} \bmod 2^{e+2}$. Using the definitions of Φ_1 and Φ_2 from Proposition 2, we obtain the explicit expressions:

$$\begin{aligned} S_1 &= (-[\alpha]Q, \mathbf{0}, \mathbf{0}, \mathbf{0}), & T_1 &= ([N_1]P, \mathbf{0}, \gamma_1(P), -\widehat{\gamma}_2(P)), \\ S_2 &= (\mathbf{0}, -[\alpha]Q, \mathbf{0}, \mathbf{0}), & T_2 &= (\mathbf{0}, [N_1]P, \gamma_2(P), \widehat{\gamma}_1(P)), \\ S_3 &= (\mathbf{0}, \mathbf{0}, [\beta]\gamma_1(P), -[\beta]\widehat{\gamma}_2(P)), & T_3 &= ([1 - \alpha 2^e]Q, \mathbf{0}, [\alpha]\gamma_1(Q), -[\alpha]\widehat{\gamma}_2(Q)), \\ S_4 &= (\mathbf{0}, \mathbf{0}, [\beta]\gamma_2(P), [\beta]\widehat{\gamma}_1(P)), & T_4 &= (\mathbf{0}, [1 - \alpha 2^e]Q, [\alpha]\gamma_2(Q), [\alpha]\widehat{\gamma}_1(Q)). \end{aligned} \tag{9}$$

Remark 6. The expression of $\mathcal{B}$ obtained from [22, Lemma 6.4.3] ensures that this basis is symplectic. However, sign choices need to be made to express

$$\gamma_1 = \phi_{\bar{\mathbf{b}}_2} \circ \phi_{\mathbf{b}_1} + \phi_{\mathbf{c}_2} \circ \phi_{\bar{\mathbf{c}}_1} \quad \text{and} \quad \gamma_2 = \phi_{\bar{\mathbf{b}}_2} \circ \phi_{\mathbf{c}_1} - \phi_{\mathbf{c}_2} \circ \phi_{\bar{\mathbf{b}}_1},$$

since the endomorphisms $\phi_{\bar{\mathbf{b}}_2} \circ \phi_{\mathbf{b}_1}$, $\phi_{\mathbf{c}_2} \circ \phi_{\bar{\mathbf{c}}_1}$, $\phi_{\bar{\mathbf{b}}_2} \circ \phi_{\mathbf{c}_1}$ and $\phi_{\mathbf{c}_2} \circ \phi_{\bar{\mathbf{b}}_1}$ are determined by $\mathbf{b}_1\bar{\mathbf{b}}_2$, $\bar{\mathbf{c}}_1\mathbf{c}_2$, $\mathbf{c}_1\bar{\mathbf{b}}_2$ and $\bar{\mathbf{b}}_1\mathbf{c}_2$ only up to sign. By assumption, for Proposition 2 to hold, these sign choices should be made consistently to ensure that isogeny diagrams from Proposition 1 commute $\widehat{\phi_{\mathbf{c}_i}} \circ \phi_{\mathbf{b}_i} = \widehat{\phi_{\bar{\mathbf{b}}_i}} \circ \phi_{\bar{\mathbf{c}}_i}$ for $i \in \{1, 2\}$.

However, for $\mathcal{B}$ to be symplectic, we only have to check a weaker condition. From Equation (9), we obtain that:

$$e_{2^{e+2}}(T_1, T_3) = e_{2^{e+2}}(T_2, T_4) = \zeta^{N_1 - 2^e + \alpha(\deg(\gamma_1) + \deg(\gamma_2))},$$

so γ_1 and γ_2 need to satisfy $N_1 - 2^e + \alpha(\deg(\gamma_1) + \deg(\gamma_2)) \equiv 0 \bmod 2^{e+2}$. By computing $\deg(\gamma_1)$ and $\deg(\gamma_2)$, we get that this condition is equivalent to:

$$\text{tr}((\phi_{\bar{\mathbf{b}}_2} \circ \phi_{\mathbf{b}_1}) \circ (\widehat{\phi_{\mathbf{c}_2} \circ \phi_{\bar{\mathbf{c}}_1}})) = \text{tr}((\phi_{\bar{\mathbf{b}}_2} \circ \phi_{\mathbf{c}_1}) \circ (\widehat{\phi_{\mathbf{c}_2} \circ \phi_{\bar{\mathbf{b}}_1}})).$$

Which by Equation (8), is equivalent to:

$$\begin{aligned} 2(B_1 D_1 + B_2 D_2 \mathbf{n}(\omega)) + \text{tr}(\omega)(B_1 D_2 + B_2 D_1) \\ = 2(C_1 E_1 + C_2 E_2 \mathbf{n}(\omega)) + \text{tr}(\omega)(C_1 E_2 + C_2 E_1). \end{aligned} \quad (10)$$

This condition follows from the commutativity of isogeny diagrams ($\widehat{\phi}_{\mathbf{c}_i} \circ \phi_{\mathbf{b}_i} = \widehat{\phi}_{\overline{\mathbf{b}_i}} \circ \phi_{\overline{\mathbf{c}_i}}$ for $i \in \{1, 2\}$) and is a sufficient condition for $\mathcal{B}$ to be symplectic. However, the converse does not hold, *i.e.* Equation (10) is not a sufficient condition for isogeny diagrams to commute.

From the basis $\mathcal{B}$ given by Equation (9), we can extract the level 2 product theta structure on the codomain $E_{\mathbf{a}} \times E_{\overline{\mathbf{a}}} \times A$ by using the following lemma to express the change of basis matrix from the symplectic basis $F_*(\mathcal{B})$ induced by F and $\mathcal{B}$ to a product symplectic basis $\mathcal{C}_0$. We can then compute the associated change of theta structure by [22, Theorem 6.2.10]. Note that to recover the product theta structure on the codomain, we need to compute the change of basis matrix from $F_*(\mathcal{B})$ to $\mathcal{C}_0$, so the inverse of the matrix given by the following lemma.

Lemma 3. *Consider the product ζ^{2^e} -symplectic basis of $(E_{\mathbf{a}} \times E_{\overline{\mathbf{a}}} \times A)[4]$ given by:*

$$\begin{aligned} \mathcal{C}_0 := [2^e]((\phi_{\mathbf{b}_1}(P), \mathbf{0}, \mathbf{0}_A), (\mathbf{0}, \phi_{\overline{\mathbf{b}_1}}(P), \mathbf{0}_A), (\mathbf{0}, \mathbf{0}, \Psi_1(P, \mathbf{0})), (\mathbf{0}, \mathbf{0}, \Psi_1(\mathbf{0}, P)), \\ ([t]\phi_{\mathbf{b}_1}(Q), \mathbf{0}, \mathbf{0}_A), (\mathbf{0}, [t]\phi_{\overline{\mathbf{b}_1}}(Q), \mathbf{0}_A), (\mathbf{0}, \mathbf{0}, [\beta]\Psi_1(Q, \mathbf{0})), (\mathbf{0}, \mathbf{0}, [\beta]\Psi_1(\mathbf{0}, Q))), \end{aligned}$$

with $t \equiv \mathbf{n}(\mathbf{b}_1)^{-1} \pmod{2^{e+2}}$. Then the change of basis matrix from $\mathcal{C}_0$ to $F_*(\mathcal{B}) := ([2^e]F(S_1), \dots, [2^e]F(S_4), F(T_1), \dots, F(T_4))$, the symplectic basis induced by $\mathcal{B}$ and F , is given (in columns) by:

$$\begin{pmatrix} 0 & -\alpha M_{1,2} & 1 & M_{1,1} & 1 & M_{1,1} & 0 & 0 \\ \alpha \overline{M}_{1,2} & 0 & -\overline{M}_{1,1} & 1 & -\overline{M}_{1,1} & 1 & 0 & 0 \\ 0 & 0 & N_1 \beta & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & N_1 \beta & 0 & 0 & 0 & 0 \\ -\alpha \mathbf{n}(\mathbf{b}_1) & -\alpha \mathbf{n}(\mathbf{b}_1) M_{2,2} & 0 & \mathbf{n}(\mathbf{b}_1) M_{2,1} & 0 & \mathbf{n}(\mathbf{b}_1) M_{2,1} & 0 & 0 \\ \alpha \mathbf{n}(\mathbf{b}_1) \overline{M}_{2,2} & -\alpha \mathbf{n}(\mathbf{b}_1) & -\mathbf{n}(\mathbf{b}_1) \overline{M}_{2,1} & 0 & -\mathbf{n}(\mathbf{b}_1) \overline{M}_{2,1} & 0 & 0 & 0 \\ \alpha N_2 & 0 & 0 & 0 & 0 & 0 & \alpha N_2 & 0 \\ 0 & \alpha N_2 & 0 & 0 & 0 & 0 & 0 & \alpha N_2 \end{pmatrix}$$

where $M, \overline{M} \in M_2(\mathbb{Z}/4\mathbb{Z})$ are respectively the change of basis matrices from $([2^e]\phi_{\mathbf{b}_1}(P), [2^e]\phi_{\mathbf{b}_1}(Q))$ to $([2^e]\phi_{\mathbf{c}_1}(P), [2^e]\phi_{\mathbf{c}_1}(Q))$ and from $([2^e]\phi_{\overline{\mathbf{b}_1}}(P), [2^e]\phi_{\overline{\mathbf{b}_1}}(Q))$ to $([2^e]\phi_{\overline{\mathbf{c}_1}}(P), [2^e]\phi_{\overline{\mathbf{c}_1}}(Q))$.

Proof. We refer to Appendix A for a proof of this computational lemma. $\square$

Once we have obtained a level 2 product theta structure $\Theta_{E_{\mathbf{a}} \times E_{\overline{\mathbf{a}}} \times A}$ on the codomain, we can extract Montgomery models for $E_{\mathbf{a}}$ and even $E_{\overline{\mathbf{a}}}$ directly from

it, following the approach from [24, Appendix B.4] that we recall here. Since the theta structure is a product, the theta null point can be writtten as:

$$\theta_{i_1, i_2, i_3, i_4}^{E_a \times E_{\bar{a}} \times A}(\mathbf{0}_{E_a \times E_{\bar{a}} \times A}) = \theta_{i_1}^{E_a}(\mathbf{0}_{E_a}) \cdot \theta_{i_2}^{E_{\bar{a}}}(\mathbf{0}_{E_{\bar{a}}}) \cdot \theta_{i_3, i_4}^A(\mathbf{0}_A),$$

for all $i_1, i_2, i_3, i_4 \in \mathbb{Z}/2\mathbb{Z}$. We can find $i_3, i_4 \in \mathbb{Z}/2\mathbb{Z}$ such that $\theta_{0,0,i_3,i_4}^{E_a \times E_{\bar{a}} \times A}(\mathbf{0}_{E_a \times E_{\bar{a}} \times A}) \neq 0$ and set $a := \theta_{0,0,i_3,i_4}^{E_a \times E_{\bar{a}} \times A}(\mathbf{0}_{E_a \times E_{\bar{a}} \times A})$, $b := \theta_{1,0,i_3,i_4}^{E_a \times E_{\bar{a}} \times A}(\mathbf{0}_{E_a \times E_{\bar{a}} \times A})$ and $b' := \theta_{0,1,i_3,i_4}^{E_a \times E_{\bar{a}} \times A}(\mathbf{0}_{E_a \times E_{\bar{a}} \times A})$. Then $(a : b)$ and $(a : b')$ are theta null points of E_a and $E_{\bar{a}}$ respectively. We can then recover the Montgomery coefficient A_a of E_a , determining its equation $y^2 = x^3 + A_a x^2 + x$ as follows:

$$A_a = \pm 2 \frac{a^4 + b^4}{a^4 - b^4},$$

where the the sign determines if we obtain the equation of E_a or its quadratic twist E_a^t . In practice, we impose that $A_a + 2$ is not a square to lift this ambiguity. We proceed similarly for $E_{\bar{a}}$.

In the CSURF context. As in Pegasus [24], we generate a basis (P, Q) of $E[2^{e+2}]$ such that $\pi(P) = P$ and $\pi(Q) = -Q$, where we recall that $e := f - 3$ and $f = v_2(p + 1)$ is the 2-adic valuation of $p + 1$. In the CSURF context, such a basis exists and we have $P \in E(\mathbb{F}_p)$ and Q can be seen as an $\mathbb{F}_p$ -rational point on the quadratic twist E^t . Hence, the $(x : z)$ -coordinates of P and Q are $\mathbb{F}_p$ -rational, which allows for a big computational gain compared to $\mathbb{F}_{p^2}$ -arithmetic.

If we denote by $\iota : R \hookrightarrow \text{End}(E)$, $\sqrt{-p} \mapsto \pi$, an orientation on E induced by the Frobenius, then $T_P := \iota(\omega)(P) - P$ and $T_Q := \iota(\omega)(Q)$ are 2-torsion points that can be computed with [24, Algorithm 4]. Recall that by Equation (8), we have:

$$\begin{aligned} \gamma_1 &:= \phi_{\bar{\mathbf{b}}_2} \circ \phi_{\mathbf{b}_1} + \phi_{\mathbf{c}_2} \circ \phi_{\bar{\mathbf{c}}_1} = [B_1 + D_1] + [B_2 + D_2]\iota(\omega), \\ \gamma_2 &:= \phi_{\mathbf{c}_2} \circ \phi_{\bar{\mathbf{b}}_1} - \phi_{\bar{\mathbf{b}}_2} \circ \phi_{\mathbf{c}_1} = [C_1 - E_1] + [C_2 - E_2]\iota(\omega). \end{aligned}$$

Furthermore, recall that $\widehat{\iota(\omega)} = (1 - \pi)/2 = 1 - \iota(\omega)$, $\alpha \equiv N_1^{-1} \pmod{2^{e+2}}$, $\beta \equiv N_2^{-1} \pmod{2^{e+2}}$, and $t \equiv \mathbf{n}(\mathbf{b}_1)^{-1} \pmod{2^{e+2}}$. The symplectic basis $\mathcal{B}$ adapted to F defined by Equation (9) can then be explicitly expressed as:

$$\begin{aligned} S_1 &= (-[\alpha]Q, \mathbf{0}, \mathbf{0}, \mathbf{0}) \\ S_2 &= (\mathbf{0}, -[\alpha]Q, \mathbf{0}, \mathbf{0}) \\ S_3 &= (\mathbf{0}, \mathbf{0}, [\beta\sigma_3]P + [\beta\sigma_6]T_P, -[\beta\sigma_1]P + [\beta\sigma_5]T_P) \\ S_4 &= (\mathbf{0}, \mathbf{0}, [\beta\sigma_2]P + [\beta\sigma_5]T_P, [\beta\sigma_4]P - [\beta\sigma_6]T_P) \\ T_1 &= ([N_1]P, \mathbf{0}, [\sigma_3]P + [\sigma_6]T_P, -[\sigma_1]P + [\sigma_5]T_P) \\ T_2 &= (\mathbf{0}, [N_1]P, [\sigma_2]P + [\sigma_5]T_P, [\sigma_4]P - [\sigma_6]T_P) \\ T_3 &= ([1 - \alpha 2^e]Q, \mathbf{0}, [\alpha\sigma_4]Q + [\alpha\sigma_6]T_Q, -[\alpha\sigma_2]Q + [\alpha\sigma_5]T_Q) \\ T_4 &= (\mathbf{0}, [1 - \alpha 2^e]Q, [\alpha\sigma_1]Q + [\alpha\sigma_5]T_Q, [\alpha\sigma_3]Q - [\alpha\sigma_6]T_Q). \end{aligned} \tag{11}$$

where:

$$\begin{aligned} \sigma_1 &:= C_1 - E_1, & \sigma_2 &:= C_1 - E_1 + C_2 - E_2, & \sigma_3 &:= B_1 + D_1 + B_2 + D_2, \\ \sigma_4 &:= B_1 + D_1, & \sigma_5 &:= C_2 - E_2, & \sigma_6 &:= B_2 + D_2. \end{aligned} \quad (12)$$

We now express the change of basis from $F_*(\mathcal{B})$ to $\mathcal{C}_0$ on the codomain $E_a \times E_{\bar{a}} \times A$ of F given by Lemma 3. Indeed, we have:

$$\begin{aligned} \phi_{c_1} &= \frac{1}{n(\mathbf{b}_1)} \phi_{b_1} \circ \phi_{\bar{b}_1 c_1} = \frac{1}{n(\mathbf{b}_1)} \phi_{b_1} \circ \widehat{\phi}_{\bar{c}_1 b_1} = \frac{1}{n(\mathbf{b}_1)} \phi_{b_1} \circ ([A_1 + A_2] - [A_2]\iota(\omega)) \\ \phi_{\bar{c}_1} &= \frac{1}{n(\mathbf{b}_1)} \phi_{\bar{b}_1} \circ \phi_{\bar{c}_1 b_1} = \frac{1}{n(\mathbf{b}_1)} \phi_{\bar{b}_1} \circ ([A_1] + [A_2]\iota(\omega)). \end{aligned}$$

Since T_P and T_Q are 2-torsion points, it follows that the change of basis matrices from $([2^e]\phi_{b_1}(P), [2^e]\phi_{b_1}(Q))$ to $([2^e]\phi_{c_1}(P), [2^e]\phi_{c_1}(Q))$ and from $([2^e]\phi_{\bar{b}_1}(P), [2^e]\phi_{\bar{b}_1}(Q))$ to $([2^e]\phi_{\bar{c}_1}(P), [2^e]\phi_{\bar{c}_1}(Q))$ are respectively given by :

$$M = \text{Diag}(tA_1, t(A_1 + A_2)) \quad \text{and} \quad \bar{M} = \text{Diag}(t(A_1 + A_2), tA_1).$$

Substituting these values into the matrix from Lemma 3 and computing its inverse, we obtain the change of basis matrix from $F_*(\mathcal{B})$ to $\mathcal{C}_0$:

$$\begin{pmatrix} 0 & 0 & 0 & 0 & -n(\mathbf{b}_1)N_1\mu & A_3N_1\mu & 0 & 0 \\ 0 & 0 & 0 & 0 & -A_1N_1\mu & -n(\mathbf{b}_1)N_1\mu & 0 & 0 \\ 0 & 0 & \alpha N_2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \alpha N_2 & 0 & 0 & 0 & 0 \\ \nu & -tA_1\nu & -\alpha N_2 & 0 & 0 & 0 & 0 & 0 \\ tA_3\nu & \nu & 0 & -\alpha N_2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & n(\mathbf{b}_1)N_1\mu & -A_3N_1\mu & \beta N_1 & 0 \\ 0 & 0 & 0 & 0 & A_1N_1\mu & n(\mathbf{b}_1)N_1\mu & 0 & \beta N_1 \end{pmatrix}$$

where $A_3 \equiv A_1 + A_2 \pmod{4}$, $\mu \equiv (n(\mathbf{b}_1)^2 + A_1A_3)^{-1} \pmod{4}$ and $\nu \equiv (1 + t^2A_1A_3)^{-1} \pmod{4}$. Using this matrix and [22, Theorem 6.2.10], we recover the desired codomain product theta structure.

4.2 Singular cases

We may encounter singular cases when computing the two first 2-isogenies f_1, f_2 of the 4-dimensional 2-isogeny chain F . These cases should be treated separately with extra care, increasing the code complexity and making a constant time implementation more difficult. For these reasons, we simply reject them.

Incomplete gluing. One singular case to avoid is when the first 2-isogeny $f_1 : E^4 \rightarrow \mathcal{A}_1$ is block diagonal and maps to a product of abelian surfaces. Similar issues have been analyzed in detail in previous works [21, 24] but we chose to avoid them for code and algorithmic simplicity. This also allows cost savings (see Section 5.1). This singularity is general and not specific to the CSURF context.

Lemma 4. *Recall the coefficients introduced in Equation (8). If $B_1 + D_1 \equiv B_2 + D_2 \equiv 0 \pmod{2}$ or $E_1 - C_1 \equiv E_2 - C_2 \equiv 0 \pmod{2}$, then the codomain $\mathcal{A}_1$ of the first 2-isogeny $f_1 : E^4 \rightarrow \mathcal{A}_1$ of the chain F is a product of abelian surfaces or elliptic curves.*

Proof. Let $\iota : R \hookrightarrow \text{End}(E)$ be the orientation of the starting curve E . Then by Proposition 2, we have:

$$\ker(F) = \{([N_1]P, [N_1]Q, \gamma_1(P) + \gamma_2(Q), \hat{\gamma}_1(Q) - \hat{\gamma}_2(P)) \mid (P, Q) \in (E \times E)[2^e]\},$$

where by Equation (8):

$$\begin{aligned}\gamma_1 &:= \phi_{\bar{b}_2} \circ \phi_{b_1} + \phi_{c_2} \circ \phi_{c_1} = [B_1 + D_1] + [B_2 + D_2]\iota(\omega), \\ \gamma_2 &:= \phi_{c_2} \circ \phi_{\bar{b}_1} - \phi_{\bar{b}_2} \circ \phi_{c_1} = [C_1 - E_1] + [C_2 - E_2]\iota(\omega).\end{aligned}$$

When $B_1 + D_1 \equiv B_2 + D_2 \equiv 0 \pmod{2}$, it follows that $\gamma_1 \equiv 0 \pmod{2}$, so that:

$$\ker(F)[2] = \{([N_1]P, [N_1]Q, \gamma_2(Q), -\hat{\gamma}_2(P)) \mid (P, Q) \in (E \times E)[2]\},$$

So f_1 is of the form $(R, S, T, U) \mapsto (\varphi_1(R, U), \varphi_2(S, T))$, where $\varphi_1 : E^2 \rightarrow \mathcal{S}_1$ and $\varphi_2 : E^2 \rightarrow \mathcal{S}_2$ are 2-dimensional 2-isogenies with kernels:

$$\ker(\varphi_1) = \{(P, -\hat{\gamma}_2(P)) \mid P \in E[2]\}, \quad \ker(\varphi_2) = \{(P, \gamma_2(P)) \mid P \in E[2]\},$$

since N_1 is odd. When furthermore, $E_1 - C_1 \equiv 1 \pmod{2}$ and $E_2 - C_2 \equiv 0 \pmod{2}$, we have $\mathcal{S}_1 \simeq \mathcal{S}_2 \simeq E^2$ and φ_1 and φ_2 both are the 2-isogeny $(R, S) \mapsto (R + S, R - S)$ up to post-composition by isomorphisms.

Similarly, in the case $E_1 - C_1 \equiv E_2 - C_2 \equiv 0 \pmod{2}$, $\gamma_2 \equiv 0 \pmod{2}$ and we can write f_1 in the form $(R, S, T, U) \mapsto (\varphi_1(R, T), \varphi_2(S, U))$. And when furthermore, $B_1 + D_1 \equiv 1 \pmod{2}$ and $B_2 + D_2 \equiv 0 \pmod{2}$, φ_1 and φ_2 are also the 2-isogeny $(R, S) \in E^2 \mapsto (R + S, R - S) \in E^2$ up to post-composition by isomorphisms. $\square$

Non-gluing singularities Even when $f_1 : E^4 \rightarrow \mathcal{A}_1$ glues perfectly, meaning that $\mathcal{A}_1$ is not a product, the codomain $\mathcal{A}_2$ of the second 2-isogeny of the chain $f_2 : \mathcal{A}_1 \rightarrow \mathcal{A}_2$ may have one zero dual theta constant which impacts its evaluation. This phenomenon is unexpected and contradicts [22, Conjecture 6.1.15.(i)], expecting that dual theta constants of an abelian variety only vanish when it is 2-isogenous to a product. Fortunately, we can predict when this singularity appears to avoid it at the norm equation phase (see Section 4.3). Note that we only observed and analysed this singularity in the CSURF context.

Lemma 5. *Assume that we work in the CSURF context, so that $\text{End}_{\mathbb{F}_p}(E) = \mathbb{Z}[(1 + \pi)/2]$. Recall the notations from Equation (12) and assume the conditions of Lemma 4 do not apply i.e. that both pairs $\{\sigma_1, \sigma_2\}$ and $\{\sigma_3, \sigma_4\}$ contain an odd element. Then, the codomain $\mathcal{A}_2$ of the second isogeny has one zero theta constant exactly when either one of $\sigma_1, \sigma_2, \sigma_3$ or σ_4 is $2 \pmod{4}$.*

Proof. We refer to Appendix B for the proof of this technical lemma. $\square$

Another singularity preventing the computation of the second isogeny $f_2 : \mathcal{A}_1 \rightarrow \mathcal{A}_2$. The second isogeny being a generic isogeny, we expect to be able to compute it from the 8-torsion points $[2^{e-2}]f_1(T_1), \dots, [2^{e-2}]f_1(T_4)$, using [22, Algorithm 6.5]. This algorithm exploits the following formula [22, Lemma 6.1.4]:

$$U_{i+e_l}^{\mathcal{A}_2}(\mathbf{0}_{\mathcal{A}_2}) \cdot H((\theta_j^{\mathcal{A}_1}([2^{e-2}]T_l)^2)_j)_i = U_i^{\mathcal{A}_2}(\mathbf{0}_{\mathcal{A}_2}) \cdot H((\theta_j^{\mathcal{A}_1}([2^{e-2}]T_l)^2)_j)_{i+e_l},$$

for all $l \in \llbracket 1 ; 4 \rrbracket$ and $i \in (\mathbb{Z}/2\mathbb{Z})^4$, where the $(U_i^{\mathcal{A}_2})_i$ are the dual theta coordinates on $\mathcal{A}_2$, H is the Hadamard transform given as a matrix by $H = ((-1)^{\langle i|j \rangle})_{i,j \in (\mathbb{Z}/2\mathbb{Z})^g}$ and e_l is the vector of $(\mathbb{Z}/2\mathbb{Z})^g$ with 1 at index l and 0 everywhere else. From this formula, we are able to retrieve the dual theta null point $(U_i^{\mathcal{A}_2}(\mathbf{0}_{\mathcal{A}_2}))_i$. However, when too many values $H((\theta_j^{\mathcal{A}_1}([2^{e-2}]T_l)^2)_j)_i$ vanish, we are not able to recover all the theta constants $U_i^{\mathcal{A}_2}(\mathbf{0}_{\mathcal{A}_2})$. We expect this to happen if f_2 is a gluing but unexpectedly, this phenomenon happens even when this is not the case. Unlike other singularities described above, we cannot predict it easily depending on the values of the coefficients from Equation (8). However, computing $[2^{e-2}]f_1(T_1 + T_4)$ and giving this additional data to the codomain computation algorithm [22, Algorithm 6.5] is a sufficient countermeasure to compute f_2 successfully in all cases.

4.3 Early rejections

If we encounter one of the cases described in Section 4.2, we must reject it and run the algorithm again. It turns out that we can predict whether we will hit a singular case before running Cornacchia, to reject those cases earlier.

Assuming N odd. Recall that $N_1 = n(\mathbf{b}_1) + n(\mathbf{c}_1)$ must be odd because it must be coprime with $N_2 = 2^e - N_1$ (see Proposition 2). If we look at $\alpha = \omega + \lambda$, its norm is $n(\alpha) = \lambda(1 + \lambda) + (p + 1)/4$, which is clearly even. Writing $rN = n(\alpha)$, and assuming N odd, we can now look at $n(\mathbf{c}_1)$. We have that $n(\mathbf{c}_1) = n(\delta_1)/N = c_1^2 N + r + c_1 \operatorname{tr}(\alpha) \equiv 0 \pmod{2}$ since $\operatorname{tr}(\alpha)$ is odd⁵, thus we require that $n(\mathbf{b}_1)$ is odd, or equivalently that b_1 is odd.

Let us denote by z the right side of Equation (7) that we have to write as a sum of squares $b_1^2 + b_2^2$ by Cornacchia's algorithm. We have 3 cases:

- if $z \equiv 0 \pmod{4}$, both b_1, b_2 will be even; we must discard this case
- if $z \equiv 1 \pmod{4}$, one of b_1, b_2 will be odd; we can fix it to be b_1 and go forward
- if $z \equiv 2 \pmod{4}$, b_1, b_2 are both odd, so there is no problem

Notice that this fixes already the parity of $n(\mathbf{c}_2)$, e.g. if $z \equiv 2 \pmod{4}$ the above reasoning together with the norm equation implies that $n(\mathbf{c}_2)$ even.

Assuming $N \equiv 2 \pmod{4}$. In this case, the conditions coming from Lemma 4 and Lemma 5 will never be satisfied simultaneously. However, we easily reduce to the case of N being odd, simply by acting with the 2-isogeny corresponding to the ideal $(2, \alpha)$ directly, and then proceeding with the ideal $(N/2, \alpha)$.

⁵This follows from $\operatorname{tr}(\alpha)^2 - 4n(\alpha) = \Delta_R = -p$.

Assuming $N \equiv 0 \pmod{4}$. In this case, we could again act directly with the 2-power part of $\mathfrak{a}$. However, this will potentially get quite costly, especially in the case of a constant time implementation. Thus, we instead choose to deal with this case directly. We must again ensure that N_i is odd. However, this time, it is $n(\mathfrak{b}_i) = b_i^2 N$ that is guaranteed to be even, and we must instead sample (c_1, c_2) until $n(\mathfrak{c}_i)$ are odd, i.e. until c_i has the opposite parity to r . Once this is achieved, we may proceed as in the odd case, by swapping the roles of $\mathfrak{b}_i$ and $\mathfrak{c}_i$.

Avoiding the conditions of Lemma 4. We can apply a similar reasoning to the other conditions described in Section 4.2. First of all, by Lemma 4, we have to avoid cases in which $\sigma_4 \equiv \sigma_6 \equiv 0 \pmod{2}$ or $\sigma_1 \equiv \sigma_5 \equiv 0 \pmod{2}$ where we recall $\sigma_4 = B_1 + D_1, \sigma_6 = B_2 + D_2, \sigma_1 = C_1 - E_1, \sigma_5 = C_2 - E_2$ as introduced in Equation (12). By construction, $B_2 = 0$, and D_1 and D_2 are fully known before running Cornacchia; the first condition can then be checked immediately. The C_i and E_i are not, but we can define $\tilde{C}_1 + \tilde{C}_2\omega = \tilde{\delta}_2, \tilde{E}_1 + \tilde{E}_2\omega = \tilde{\delta}_1$ which are known, and such that $b_1\tilde{C}_i = C_i$ and $b_2\tilde{E}_i = E_i$. Since the values of the $b_i \pmod{2}$ are known before Cornacchia, as shown above, the condition of Lemma 4 can also be tested.

Avoiding the conditions of Lemma 5. Lastly, according to Lemma 5, we also need to avoid cases in which one of the quantities

$$\sigma_4 = B_1 + D_1, \quad \sigma_3 = B_1 + D_1 + D_2, \sigma_1 = C_1 - E_1, \quad \sigma_2 = C_1 + C_2 - E_1 - E_2$$

is $2 \pmod{4}$. To do so we need to determine the value of $b_i \pmod{4}$, by looking at the value of $z \pmod{8}$:

- if $z \equiv 1 \pmod{8}$, then $b_2 \equiv 0 \pmod{4}$. Since b_2 divides E_1, E_2 and B_1 , all those terms will be $0 \pmod{4}$. We have to reject cases in which $D_1, D_1 + D_2, \tilde{C}_1$ or $\tilde{C}_1 + \tilde{C}_2$ are $2 \pmod{4}$ (notice that b_1 is always odd);
- if $z \equiv 5 \pmod{8}$, the situation is analogous, but $b_2 \equiv 2 \pmod{4}$. We must then check for $D_1 \equiv 0 \pmod{4}, D_1 + D_2 \equiv 0 \pmod{4}, \tilde{C}_1 - 2\tilde{E}_1 \equiv 2 \pmod{4}$ and $\tilde{C}_1 + \tilde{C}_2 - 2\tilde{E}_1 - 2\tilde{E}_2 \equiv 2 \pmod{4}$;
- if $z \equiv 2 \pmod{8}$ the situation is a bit different: here both b_1 and b_2 are odd, which means $\pm 1 \pmod{4}$, but we are free to replace either of them with its negative after running Cornacchia. Notice that if we replace both b_1 with $-b_1$ and b_2 with $-b_2$ all the σ_i stay the same; we thus restrict ourselves to consider the cases $(b_1, b_2) \equiv (1, 1) \pmod{4}$ and $(b_1, b_2) \equiv (1, -1) \pmod{4}$. In the first case, we have $B_1 \equiv N \pmod{4}, C_i \equiv \tilde{C}_i \pmod{4}$ and $E_i \equiv \tilde{E}_i \pmod{4}$, while in the second case the sign of B_1 and the E_i is swapped. We can then check the singular conditions directly, and run Cornacchia only on the instances in which one of these two cases does not hit a singularity. The sign of b_1 and b_2 has then to be updated accordingly.

Remark 7. Note that changing the sign of $b_1 \pmod{4}$ does not help if z is 1 or $5 \pmod{4}$, since all we can do is change the sign of quantities that are already even $\pmod{4}$. Moreover, the choices of b_1 and b_2 in Remark 4 need to be consistent following Equation (10).

5 Implementation

We provide an implementation of our algorithm in SageMath 10.5. The code is public, and can be found at

<https://github.com/KULeuven-COSIC/qt-pegasis>

5.1 Timings

We timed our implementation by averaging 100 runs on an Intel Core i5-1235U CPU. The results are reported in Table 1, compared with Pegasis [24], and run on the same machine and with the same parameters. In both cases, Step 1 refers to finding a solution to the norm equation, Step 2 is the derivation of the kernel and Step 3 is the computation of the 4-dimensional isogenies.

As we can see, qt-Pegasis outperforms Pegasis in all three steps of the algorithm. This can be attributed to the following reasons:

Step 1. In Step 1, qt-Pegasis replaces the old FindUV algorithm from Pegasis [24, Algorithm 2] with Algorithm 1, which is significantly faster for a number of reasons. Perhaps the clearest reason is that the most costly subroutine in both cases is the repeated calls to Cornacchia until a solution is found. However, in Pegasis, this required two random numbers to be sums of squares at the same time, while for qt-Pegasis, it is only a single one.

Step 2. The biggest difference in performance overall comes from Step 2, which, in Pegasis, involved the expensive computation of Elkies isogenies. In qt-Pegasis, this is now simply replaced by a few point multiplications.

Step 3. Finally, we see that the performance of Step 3 remains similar for both Pegasis and qt-Pegasis. On the one hand, our 4D chains are on average a few steps longer than in Pegasis (due to Algorithm 1 always solving the norm equation for a fixed 2^e), but on the other hand, we now avoid all cases where the chain does not glue immediately, which are more costly in Pegasis.

5.2 Towards an optimized implementation

Our current implementation relies on SageMath, and focuses on showing the correctness and general feasibility of our approach. Various algorithmic and implementation improvements are possible when moving to a lower level language such as C or Rust, and for specific protocols, e.g. where we need to act on the same curve.

Table 2 shows a more detailed breakdown of the timings of qt-Pegasis. In particular, Step 2 is split in 2.1, in which a suitable basis for the 2^{e+2} -torsion is computed, and 2.2, in which the kernel points T_i are computed from such basis. Step 1 can be precomputed when acting multiple times with the same ideal, while Step 2.1 can be precomputed when acting on the same curve. Step 2.2 is dominated by multiplying 2 points by 5 scalars each, which can be batched.

Step 3 is by far the most expensive, taking between 91% and 95% of the total computational time at all levels. This step can be implemented fully using

$\lceil \log_2(p) \rceil$	Prime p	Variant	Step 1	Step 2	Step 3	Total	Improvement
508	$3 \cdot 11 \cdot 2^{503} - 1$	Pegasis	0.097	0.48	0.96	1.53	$1.8\times$
		qt-Pegasis	0.006	0.05	0.79	0.85	
1008	$3 \cdot 5 \cdot 2^{1004} - 1$	Pegasis	0.21	1.16	2.84	4.21	$1.7\times$
		qt-Pegasis	0.014	0.18	2.29	2.48	
1554	$3^2 \cdot 2^{1551} - 1$	Pegasis	1.19	2.85	6.49	10.5	$1.9\times$
		qt-Pegasis	0.008	0.423	5.10	5.54	
2031	$3 \cdot 17 \cdot 2^{2026} - 1$	Pegasis	1.68	8.34	11.3	21.3	$2.2\times$
		qt-Pegasis	0.06	0.76	8.87	9.69	
4089	$3^2 \cdot 7 \cdot 2^{4084} - 1$	Pegasis	15.6	52.8	53.5	122	$2.6\times$
		qt-Pegasis	0.67	3.83	43.1	47.6	

Table 1. Comparison of timings (in s) between Pegasis and qt-Pegasis

optimized arithmetic for the finite field $\mathbb{F}_p$, resulting in a large speedup. Any other improvement to 4-dimensional isogeny computations would also result in a significant speedup for qt-Pegasis, since this is the dominating step.

Prime size	T1	T2.1	T2.2	T3	Tot	T3 %
508	0.006	0.013	0.038	0.789	0.845	93%
1008	0.014	0.043	0.134	2.290	2.418	95%
1554	0.008	0.105	0.322	5.105	5.538	92%
2031	0.058	0.188	0.572	8.866	9.685	92%
4089	0.672	0.980	2.853	43.11	47.62	91%

Table 2. Timings breakdown for qt-Pegasis. T1: solve norm equation. T2.1: basis generation of 2^{e+2} torsion. T2.2: kernel points. T3: chain of 4d isogenies. The last column shows the percentage of time spent in T3 compared to the total time.

The overall simplicity of our algorithm, especially the fact that we can avoid auxiliary isogenies computed using Elkies' algorithm, implies further benefits compared to Pegasis. First of all, we are not constrained to pick the base field $\mathbb{F}_p$ so that small primes are split in $\mathbb{Z}[\sqrt{-p}]$; this gives us freedom to choose primes with a smaller cofactor, or faster arithmetic. We include some examples in our implementation. Moreover, qt-Pegasis is much more suitable for a constant time implementation. The biggest obstacles seem to be Cornacchia's algorithm in Step 1. Note however that this step accounts for a relatively small part of the overall computational time; we therefore expect the overhead of a constant time implementation to be rather small.

References

- [1] M. A. Aardal, G. Adj, D. F. Aranha, A. Basso, I. A. Canales Martínez, J. Chávez-Saab, M. C. Santos, P. Dartois, L. De Feo, M. Duparc, J. K. Eriksen, T. B. Fouotsa, D. L. G. Filho, B. Hess, D. Kohel, A. Leroux, P. Longa, L. Maino, M. Meyer, K. Nakagawa, H. Onuki, L. Panny, S. Patranabis, C. Petit, G. Pope, K. Reijnders, D. Robert, F. Rodríguez Henríquez, S. Schaeffler, and B. Wesolowski. *SQIsign*. Tech. rep. available at <https://csrc.nist.gov/Projects/pqc-dig-sig/round-2-additional-signatures>. National Institute of Standards and Technology, 2024.
- [2] M. A. Aardal, A. Basso, L. De Feo, S. Patranabis, and B. Wesolowski. *A Complete Security Proof of SQIsign*. Cryptology ePrint Archive, Report 2025/379. 2025.
- [3] B. Allombert, J.-F. Biasse, J. K. Eriksen, P. Kutas, C. Leonardi, A. Page, R. Scheidler, and M. T. Bagi. “Faster SCALLOP from Non-prime Conductor Suborders in Medium Sized Quadratic Fields”. In: *PKC 2025, Part III*. Ed. by T. Jager and J. Pan. Vol. 15676. LNCS. Springer, Cham, May 2025, pp. 333–363.
- [4] A. Basso, G. Borin, W. Castryck, M. Corte-Real Santos, R. Invernizzi, A. Leroux, L. Maino, F. Vercauteren, and B. Wesolowski. “PRISM: Simple and Compact Identification and Signatures from Large Prime Degree Isogenies”. In: *PKC 2025, Part III*. Ed. by T. Jager and J. Pan. Vol. 15676. LNCS. Springer, Cham, May 2025, pp. 300–332.
- [5] A. Basso, P. Dartois, L. De Feo, A. Leroux, L. Maino, G. Pope, D. Robert, and B. Wesolowski. “SQIsign2D-West - The Fast, the Small, and the Safer”. In: *ASIACRYPT 2024, Part III*. Ed. by K.-M. Chung and Y. Sasaki. Vol. 15486. LNCS. Springer, Singapore, Dec. 2024, pp. 339–370.
- [6] J. V. Belding. “Number theoretic algorithms for elliptic curves”. PhD thesis. PhD thesis, University of Maryland College Park, 2008.
- [7] D. J. Bernstein, L. De Feo, A. Leroux, and B. Smith. “Faster computation of isogenies of large prime degree”. In: *Open Book Series, Proceedings of the Fourteenth Algorithmic Number Theory Symposium – ANTS XIV 4.1* (2020), pp. 39–55.
- [8] D. J. Bernstein, T. Lange, C. Martindale, and L. Panny. “Quantum Circuits for the CSIDH: Optimizing Quantum Evaluation of Isogenies”. In: *EUROCRYPT 2019, Part II*. Ed. by Y. Ishai and V. Rijmen. Vol. 11477. LNCS. Springer, Cham, May 2019, pp. 409–441.
- [9] W. Beullens, T. Kleinjung, and F. Vercauteren. “CSI-FiSh: Efficient Isogeny Based Signatures Through Class Group Computations”. In: *ASIACRYPT 2019, Part I*. Ed. by S. D. Galbraith and S. Moriai. Vol. 11921. LNCS. Springer, Cham, Dec. 2019, pp. 227–247.
- [10] C. Birkenhake and H. Lange. *Complex Abelian Varieties*. Berlin, Heidelberg: Springer Berlin Heidelberg, 2004.
- [11] X. Bonnetain and A. Schrottenloher. “Quantum Security Analysis of CSIDH”. In: *EUROCRYPT 2020, Part II*. Ed. by A. Canteaut and Y. Ishai. Vol. 12106. LNCS. Springer, Cham, May 2020, pp. 493–522.

- [12] G. Borin, M. C.-R. Santos, J. K. Eriksen, R. Invernizzi, M. Mula, S. Schaeffler, and F. Vercauteren. “Qlapoti: Simple and Efficient Translation of Quaternion Ideals to Isogenies”. In: *Cryptology ePrint Archive* (2025).
- [13] W. Castryck and T. Decru. “CSIDH on the Surface”. In: *Post-Quantum Cryptography - 11th International Conference, PQCrypto 2020*. Ed. by J. Ding and J.-P. Tillich. Springer, Cham, 2020, pp. 111–129.
- [14] W. Castryck, T. Lange, C. Martindale, L. Panny, and J. Renes. “CSIDH: An Efficient Post-Quantum Commutative Group Action”. In: *ASIACRYPT 2018, Part III*. Ed. by T. Peyrin and S. Galbraith. Vol. 11274. LNCS. Springer, Cham, Dec. 2018, pp. 395–427.
- [15] M. Chen, A. Leroux, and L. Panny. “SCALLOP-HD: Group Action from 2-Dimensional Isogenies”. In: *PKC 2024, Part II*. Ed. by Q. Tang and V. Teague. Vol. 14603. LNCS. Springer, Cham, Apr. 2024, pp. 190–216.
- [16] L. Colò. *Oriented supersingular elliptic curves and class group actions*. https://www.leonardocolo.com/documents/thesis/PhD_Thesis.pdf. 2022.
- [17] L. Colò and D. Kohel. “Orienting supersingular isogeny graphs”. In: *Journal of Mathematical Cryptology* 14.1 (2020), pp. 414–437.
- [18] M. Corte-Real Santos, J. K. Eriksen, M. Meyer, and K. Reijnders. “AprèsSQI: Extra Fast Verification for SQIsign Using Extension-Field Signing”. In: *EUROCRYPT 2024, Part I*. Ed. by M. Joye and G. Leander. Vol. 14651. LNCS. Springer, Cham, May 2024, pp. 63–93.
- [19] C. Costello and B. Smith. “Montgomery curves and their arithmetic - The case of large characteristic fields”. In: *Journal of Cryptographic Engineering* 8.3 (Sept. 2018), pp. 227–240.
- [20] J.-M. Couveignes. *Hard Homogeneous Spaces*. Cryptology ePrint Archive, Report 2006/291. 2006.
- [21] P. Dartois. “Fast computation of 2-isogenies in dimension 4 and cryptographic applications”. In: *Journal of Algebra* 683 (2025), pp. 449–514.
- [22] P. Dartois. “Fast computation of higher dimensional isogenies for cryptographic applications”. PhD thesis. Université de Bordeaux, France, July 2025.
- [23] P. Dartois. *Towards constant time implementation of 2-isogenies in dimension 4*. In preparation. 2025.
- [24] P. Dartois, J. K. Eriksen, T. B. Fouotsa, A. Herlédan Le Merdy, R. Invernizzi, D. Robert, R. Rueger, F. Vercauteren, and B. Wesolowski. “PEGASIS: Practical Effective Class Group Action using 4-Dimensional Isogenies”. In: *Advances in Cryptology - CRYPTO 2025*. Ed. by Y. Tauman Kalai and S. F. Kamara. Cham: Springer Nature Switzerland, 2025, pp. 67–99.
- [25] P. Dartois, J. K. Eriksen, T. B. Fouotsa, A. H. L. Merdy, R. Invernizzi, D. Robert, R. Rueger, F. Vercauteren, and B. Wesolowski. *PEGASIS: Practical Effective Class Group Action using 4-Dimensional Isogenies*. Cryptology ePrint Archive, Report 2025/401. 2025.
- [26] L. De Feo, T. B. Fouotsa, P. Kutas, A. Leroux, S.-P. Merz, L. Panny, and B. Wesolowski. “SCALLOP: Scaling the CSI-FiSh”. In: *PKC 2023, Part I*.

- Ed. by A. Boldyreva and V. Kolesnikov. Vol. 13940. LNCS. Springer, Cham, May 2023, pp. 345–375.
- [27] L. De Feo, D. Kohel, A. Leroux, C. Petit, and B. Wesolowski. “SQISign: Compact Post-quantum Signatures from Quaternions and Isogenies”. In: *ASIACRYPT 2020, Part I*. Ed. by S. Moriai and H. Wang. Vol. 12491. LNCS. Springer, Cham, Dec. 2020, pp. 64–93.
 - [28] L. De Feo, A. Leroux, P. Longa, and B. Wesolowski. “New Algorithms for the Deuring Correspondence - Towards Practical and Secure SQISign Signatures”. In: *EUROCRYPT 2023, Part V*. Ed. by C. Hazay and M. Stam. Vol. 14008. LNCS. Springer, Cham, Apr. 2023, pp. 659–690.
 - [29] L. De Feo and M. Meyer. “Threshold Schemes from Isogeny Assumptions”. In: *PKC 2020, Part II*. Ed. by A. Kiayias, M. Kohlweiss, P. Wallden, and V. Zikas. Vol. 12111. LNCS. Springer, Cham, May 2020, pp. 187–212.
 - [30] M. Duparc. *Superglue: Fast formulae for $(2, 2)$ gluing isogenies*. Cryptology ePrint Archive, Paper 2025/736. 2025.
 - [31] N. D. Elkies. “Elliptic and modular curves over finite fields and related computational issues”. In: *Computational Perspectives on Number Theory: Proceedings of a Conference in Honor of A. O. L. Atkin*. Ed. by D. A. Buell and J. T. Teitelbaum. Vol. 7. AMS/IP Studies in Advanced Mathematics. American Mathematical Society, International Press, 1998, pp. 21–76.
 - [32] L. D. Feo. *Mathematics of Isogeny Based Cryptography*. 2017.
 - [33] D. Kohel, K. Lauter, C. Petit, and J.-P. Tignol. *On the quaternion ℓ -isogeny path problem*. 2014.
 - [34] G. Kuperberg. “A subexponential-time quantum algorithm for the dihedral hidden subgroup problem”. In: *SIAM Journal on Computing* 35.1 (2005), pp. 170–188.
 - [35] S. Lang. *Elliptic Functions*. Springer-Verlag, 1987, p. 324.
 - [36] A. Leroux and M. Roméas. “Updatable Encryption from Group Actions”. In: *Post-Quantum Cryptography - 15th International Workshop, PQCrypto 2024, Part II*. Ed. by M.-J. Saarinen and D. Smith-Tone. Springer, Cham, June 2024, pp. 20–53.
 - [37] J. S. Milne. “Abelian Varieties”. In: *Arithmetic Geometry*. New York, NY: Springer New York, 1986, pp. 103–150.
 - [38] T. Moriya, H. Onuki, and T. Takagi. “SiGamal: A Supersingular Isogeny-Based PKE and Its Application to a PRF”. In: *ASIACRYPT 2020, Part II*. Ed. by S. Moriai and H. Wang. Vol. 12492. LNCS. Springer, Cham, Dec. 2020, pp. 551–580.
 - [39] D. Mumford. “On the equations defining abelian varieties 1”. In: *Inventiones mathematicae* 1.4 (1966), pp. 287–354.
 - [40] D. Mumford. *Abelian varieties*. Second Edition. Tata Institute of fundamental research studies in mathematics. London: Oxford University Press, 1974, pp. x+279.
 - [41] H. Onuki. “On oriented supersingular elliptic curves”. In: *Finite Fields and Their Applications* 69 (2021), p. 101777.

- [42] A. Page and D. Robert. *Introducing Clapoti(s): Evaluating the isogeny class group action in polynomial time*. Cryptology ePrint Archive, Report 2023/1766. 2023.
- [43] L. Panny, C. Petit, and M. Stopar. *KLaPoTi: An asymptotically efficient isogeny group action from 2-dimensional isogenies*. Cryptology ePrint Archive, Report 2024/1844. 2024.
- [44] C. Peikert. “He Gives C-Sieves on the CSIDH”. In: *EUROCRYPT 2020, Part II*. Ed. by A. Canteaut and Y. Ishai. Vol. 12106. LNCS. Springer, Cham, May 2020, pp. 463–492.
- [45] D. Robert. “Theta functions and cryptographic applications”. PhD thesis. Université Henri-Poincaré, Nancy 1, France, July 2010.
- [46] D. Robert. *A note on optimising 2^n -isogenies in higher dimension*. http://www.normalesup.org/~robert/pro/publications/notes/2023-06-optimising_isogenies.pdf. 2023.
- [47] A. Rostovtsev and A. Stolbunov. *Public-Key Cryptosystem Based On Isogenies*. Cryptology ePrint Archive, Report 2006/145. 2006.
- [48] J. Vélú. “Isogénies entre courbes elliptiques”. In: *Comptes-rendus de l’Académie des Sciences* 273 (1971). Available at <https://gallica.bnf.fr>, pp. 238–241.
- [49] W. C. Waterhouse. “Abelian varieties over finite fields”. eng. In: *Annales scientifiques de l’École Normale Supérieure* 2.4 (1969), pp. 521–560.

A Splitting change of theta structure: proof Lemma 3

Lemma 3. Consider the product ζ^{2^e} -symplectic basis of $(E_{\mathfrak{a}} \times E_{\overline{\mathfrak{a}}} \times A)[4]$ given by:

$$\begin{aligned} \mathcal{C}_0 := & [2^e](\phi_{\mathfrak{b}_1}(P), \mathbf{0}, \mathbf{0}_A), (\mathbf{0}, \phi_{\overline{\mathfrak{b}_1}}(P), \mathbf{0}_A), (\mathbf{0}, \mathbf{0}, \Psi_1(P, \mathbf{0})), (\mathbf{0}, \mathbf{0}, \Psi_1(\mathbf{0}, P)) \\ & ([t]\phi_{\mathfrak{b}_1}(Q), \mathbf{0}, \mathbf{0}_A), (\mathbf{0}, [t]\phi_{\overline{\mathfrak{b}_1}}(Q), \mathbf{0}_A), (\mathbf{0}, \mathbf{0}, [\beta]\Psi_1(Q, \mathbf{0})), (\mathbf{0}, \mathbf{0}, [\beta]\Psi_1(\mathbf{0}, Q)), \end{aligned}$$

with $t \equiv \mathfrak{n}(\mathfrak{b}_1)^{-1} \pmod{2^{e+2}}$. Then the change of basis matrix from $\mathcal{C}_0$ to $F_*(\mathcal{B}) := ([2^e]F(S_1), \dots, [2^e]F(S_4), F(T_1), \dots, F(T_4))$, the symplectic basis induced by $\mathcal{B}$ and F , is given (in columns) by:

$$\begin{pmatrix} 0 & -\alpha M_{1,2} & 1 & M_{1,1} & 1 & M_{1,1} & 0 & 0 \\ \alpha \overline{M}_{1,2} & 0 & -\overline{M}_{1,1} & 1 & -\overline{M}_{1,1} & 1 & 0 & 0 \\ 0 & 0 & N_1 \beta & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & N_1 \beta & 0 & 0 & 0 & 0 \\ -\alpha \mathfrak{n}(\mathfrak{b}_1) & -\alpha \mathfrak{n}(\mathfrak{b}_1) M_{2,2} & 0 & \mathfrak{n}(\mathfrak{b}_1) M_{2,1} & 0 & \mathfrak{n}(\mathfrak{b}_1) M_{2,1} & 0 & 0 \\ \alpha \mathfrak{n}(\mathfrak{b}_1) \overline{M}_{2,2} & -\alpha \mathfrak{n}(\mathfrak{b}_1) & -\mathfrak{n}(\mathfrak{b}_1) \overline{M}_{2,1} & 0 & -\mathfrak{n}(\mathfrak{b}_1) \overline{M}_{2,1} & 0 & 0 & 0 \\ \alpha N_2 & 0 & 0 & 0 & 0 & 0 & \alpha N_2 & 0 \\ 0 & \alpha N_2 & 0 & 0 & 0 & 0 & 0 & \alpha N_2 \end{pmatrix}$$

where $M, \overline{M} \in M_2(\mathbb{Z}/4\mathbb{Z})$ are respectively the change of basis matrices from $([2^e]\phi_{\mathfrak{b}_1}(P), [2^e]\phi_{\mathfrak{b}_1}(Q))$ to $([2^e]\phi_{\mathfrak{c}_1}(P), [2^e]\phi_{\mathfrak{c}_1}(Q))$ and from $([2^e]\phi_{\overline{\mathfrak{b}_1}}(P), [2^e]\phi_{\overline{\mathfrak{b}_1}}(Q))$ to $([2^e]\phi_{\overline{\mathfrak{c}_1}}(P), [2^e]\phi_{\overline{\mathfrak{c}_1}}(Q))$.

Proof. We have:

$$\begin{aligned} F(S_1) &= F(-[\alpha]Q, \mathbf{0}, \mathbf{0}, \mathbf{0}) = (-[\alpha]\Phi_1(Q, \mathbf{0}), [\alpha]\Psi_1(Q, \mathbf{0})) \\ &= (-[\alpha]\phi_{\mathfrak{b}_1}(Q), [\alpha]\phi_{\overline{\mathfrak{c}_1}}(Q), [\alpha]\Psi_1(Q, \mathbf{0})) \\ F(S_2) &= F(\mathbf{0}, -[\alpha]Q, \mathbf{0}, \mathbf{0}) = (-[\alpha]\Phi_1(\mathbf{0}, Q), [\alpha]\Psi_1(\mathbf{0}, Q)) \\ &= (-[\alpha]\phi_{\mathfrak{c}_1}(Q), -[\alpha]\phi_{\overline{\mathfrak{b}_1}}(Q), [\alpha]\Psi_1(\mathbf{0}, Q)) \\ F(S_3) &= F(\mathbf{0}, \mathbf{0}, [\beta]\Phi_2 \circ \Phi_1(P, \mathbf{0})) = ([\beta]\tilde{\Phi}_2 \circ \Phi_2 \circ \Phi_1(P, \mathbf{0}), [\beta]\tilde{\Psi}_2 \circ \Phi_2 \circ \Phi_1(P, \mathbf{0})) \\ &= ([N_2\beta]\Phi_1(P, \mathbf{0}), [N_1\beta]\Psi_1(P, \mathbf{0})) \\ &= (\phi_{\mathfrak{b}_1}(P), -\phi_{\overline{\mathfrak{c}_1}}(P), [N_1\beta]\Psi_1(P, \mathbf{0})) \\ F(S_4) &= F(\mathbf{0}, \mathbf{0}, [\beta]\Phi_2 \circ \Phi_1(\mathbf{0}, P)) = ([N_2\beta]\Phi_1(\mathbf{0}, P), [N_1\beta]\Psi_1(\mathbf{0}, P)) \\ &= (\phi_{\mathfrak{c}_1}(P), \phi_{\overline{\mathfrak{b}_1}}(P), [N_1\beta]\Psi_1(\mathbf{0}, P)) \\ F(T_1) &= F([N_1]P, \mathbf{0}, \Phi_2 \circ \Phi_1(P, \mathbf{0})) \\ &= (\Phi_1([N_1]P, \mathbf{0}) + \tilde{\Phi}_2 \circ \Phi_2 \circ \Phi_1(P, \mathbf{0}), -\Psi_1([N_1]P, \mathbf{0}) + \tilde{\Psi}_2 \circ \Phi_2 \circ \Phi_1(P, \mathbf{0})) \\ &= ([N_1 + N_2]\Phi_1(P, \mathbf{0}), -[N_1]\Psi_1(P, \mathbf{0}) + [N_1]\Psi_1(P, \mathbf{0})) \\ &= ([2^e]\phi_{\mathfrak{b}_1}(P), -[2^e]\phi_{\overline{\mathfrak{c}_1}}(P), \mathbf{0}_A) \\ F(T_2) &= F(\mathbf{0}, [N_1]P, \Phi_2 \circ \Phi_1(\mathbf{0}, P)) \\ &= ([2^e]\Phi_1(\mathbf{0}, P), \mathbf{0}) = ([2^e]\phi_{\mathfrak{c}_1}(P), [2^e]\phi_{\overline{\mathfrak{b}_1}}(P), \mathbf{0}_A) \end{aligned}$$

$$\begin{aligned}
F(T_3) &= F([1 - \alpha 2^e]Q, \mathbf{0}, [\alpha]\Phi_2 \circ \Phi_1(Q, \mathbf{0})) \\
&= ([1 - \alpha 2^e]\Phi_1(Q, \mathbf{0}) + [\alpha]\tilde{\Phi}_2 \circ \Phi_2 \circ \Phi_1(Q, \mathbf{0}), \\
&\quad - [1 - \alpha 2^e]\Psi_1(Q, \mathbf{0}) + [\alpha]\tilde{\Psi}_2 \circ \Phi_2 \circ \Phi_1(Q, \mathbf{0})) \\
&= ([\alpha(N_1 + N_2 - 2^e)]\Phi_1(Q, \mathbf{0}), [\alpha N_1 - 1 + \alpha 2^e]\Psi_1(Q, \mathbf{0})) \\
&= (\mathbf{0}, \mathbf{0}, [\alpha 2^e]\Psi_1(Q, \mathbf{0})) \\
F(T_4) &= F(\mathbf{0}, [1 - \alpha 2^e]Q, [\alpha]\Phi_2 \circ \Phi_1(\mathbf{0}, Q)) = (\mathbf{0}, \mathbf{0}, [\alpha 2^e]\Psi_1(\mathbf{0}, Q)).
\end{aligned}$$

The result follows. $\square$

B Singularity on the second codomain: proof of Lemma 5

In this section, we prove Lemma 5 by finding conditions for a dual theta null point of the codomain $\mathcal{A}_2$ of $f_2 : \mathcal{A}_1 \rightarrow \mathcal{A}_2$ to have one vanishing coordinate. This lemma is a consequence of the following result that helps to identify zero dual theta constants.

Lemma 6. *Let $(A, \lambda_A), (C, \lambda_C)$ be principally polarised abelian varieties of dimension g defined over an algebraically closed field k (with $\text{char}(k) \neq 2$) and let $f : (A, \lambda_A) \rightarrow (C, \lambda_C)$ be a 4-isogeny between them.*

Let $\mathcal{B} := (S_1, \dots, S_g, T_1, \dots, T_g)$ be a symplectic basis of $A[16]$ adapted to f i.e. such that $\ker(f) = \langle [4]T_1, \dots, [4]T_g \rangle$. Let Θ_A be the level 2 symmetric theta structure on (A, λ_A) induced by $[4]\mathcal{B}$ and let Θ_C be the level 2 symmetric theta structure on (C, λ_C) induced by $f_(\mathcal{B}) := ([4]f(S_1), \dots, [4]f(S_g), f(T_1), \dots, f(T_g))$.*

Let $(\theta_i^A)_i, (\theta_i^C)_i$ be the theta coordinates associated to Θ_A and Θ_C respectively let and $(U_i^C)_i$ be the dual theta coordinates given by:

$$\forall i \in (\mathbb{Z}/2\mathbb{Z})^g, \quad U_i^C = \sum_{j \in (\mathbb{Z}/2\mathbb{Z})^g} (-1)^{\langle i|j \rangle} \theta_j^C,$$

where $\langle \cdot | \cdot \rangle$ is the usual scalar product. For all $i \in (\mathbb{Z}/2\mathbb{Z})^g$ (seen as $\{0, 1\}^g$), let us denote:

$$T_i := \sum_{l=1}^g [i_l]T_l.$$

Then, for all $i \in (\mathbb{Z}/2\mathbb{Z})^g$, there exists $\lambda_i \in k^*$ such that:

$$U_i^C(0_C)^2 = \lambda_i \cdot \theta_0^A([4]T_i)^2.$$

In particular, $U_i^C(0_C) = 0$ if and only if $\theta_0^A([4]T_i) = 0$.

Proof. By [22, Lemma 6.3.1], f can be decomposed into $f := f_2 \circ f_1$, where $f_1 : (A, \lambda_A) \rightarrow (B, \lambda_B)$ and $f_2 : (B, \lambda_B) \rightarrow (C, \lambda_C)$ are 2-isogenies between principally polarised abelian varieties. Then we can consider the symplectic basis $\mathcal{B}_1 := (f_1)_*(\mathcal{B}) = ([2]f_1(S_1), \dots, [2]f_1(S_g), f_1(T_1), \dots, f_1(T_g))$ of $B[8]$ and the

level 2 theta structure Θ_B induced by $[2]\mathcal{B}_1$. Then $\mathcal{B}_1$ is adapted to f_2 and Θ_C is induced by $f_*(\mathcal{B}) = (f_2)_*(\mathcal{B}_1)$. We can then apply [22, Corollary 6.1.3] to obtain that there exists $\lambda \in k^*$ such that for all $i \in (\mathbb{Z}/2\mathbb{Z})^g$,

$$U_i^C(0_C)^2 = \lambda \sum_{t \in (\mathbb{Z}/2\mathbb{Z})^g} (-1)^{\langle i|t \rangle} \theta_t^B(0_B)^2.$$

The Hadamard transform being involutive up to a 2^g factor, we obtain easily that:

$$\forall t \in (\mathbb{Z}/2\mathbb{Z})^g, \quad \theta_t^B(0_B) = \frac{1}{2^g} \sum_{j \in (\mathbb{Z}/2\mathbb{Z})^g} (-1)^{\langle j|t \rangle} U_j^B(0_B),$$

where the $(U_j^B)_j$ are the dual theta coordinates associated to Θ_B . It follows that for all $i \in (\mathbb{Z}/2\mathbb{Z})^g$,

$$\begin{aligned} U_i^C(0_C)^2 &= \frac{\lambda}{4^g} \sum_{t, v, w \in (\mathbb{Z}/2\mathbb{Z})^g} (-1)^{\langle i+v+w|t \rangle} U_v^B(0_B) U_w^B(0_B) \\ &= \frac{\lambda}{4^g} \sum_{v, w \in (\mathbb{Z}/2\mathbb{Z})^g} \left(\sum_{t \in (\mathbb{Z}/2\mathbb{Z})^g} (-1)^{\langle i+v+w|t \rangle} \right) U_v^B(0_B) U_w^B(0_B) \\ &= \frac{\lambda}{4^g} \sum_{v, w \in (\mathbb{Z}/2\mathbb{Z})^g} 2^g \delta_{i+v+w, 0} U_v^B(0_B) U_w^B(0_B) \\ &= \frac{\lambda}{2^g} \sum_{v \in (\mathbb{Z}/2\mathbb{Z})^g} U_v^B(0_B) U_{i+v}^B(0_B) \end{aligned} \tag{13}$$

Since Θ_B is induced by $[2]\mathcal{B}_1 = (f_1)_*([2]\mathcal{B})$ and $[2]\mathcal{B}$ is adapted to f_1 , [22, Eq. (5.12)] ensures that for all $i, v \in (\mathbb{Z}/2\mathbb{Z})^g$,

$$\begin{aligned} U_{i+v}^B(0_B) &= \sum_{j \in (\mathbb{Z}/2\mathbb{Z})^g} (-1)^{\langle i+v|j \rangle} \theta_j^B(0_B) = \sum_{j \in (\mathbb{Z}/2\mathbb{Z})^g} (-1)^{\langle v|j \rangle} \theta_j^B([4]f_1(T_i)) \\ &= U_v^B([4]f_1(T_i)). \end{aligned}$$

It follows by [22, Corollary 6.1.3] that for all $i \in (\mathbb{Z}/2\mathbb{Z})^g$, there exists $\mu_i \in k^*$ such that for all $v \in (\mathbb{Z}/2\mathbb{Z})^g$,

$$\begin{aligned} U_v^B(0_B) U_{i+v}^B(0_B) &= U_v^B(0_B) U_v^B(f_1([4]T_i)) \\ &= \mu_i \sum_{t \in (\mathbb{Z}/2\mathbb{Z})^g} (-1)^{\langle t|v \rangle} \theta_t^B([4]T_i)^2. \end{aligned} \tag{14}$$

Combining Equation (13) with Equation (14), we finally conclude that for all $i \in (\mathbb{Z}/2\mathbb{Z})^g$,

$$U_i^C(0_C)^2 = \frac{\lambda \mu_i}{2^g} \sum_{v \in (\mathbb{Z}/2\mathbb{Z})^g} \sum_{t \in (\mathbb{Z}/2\mathbb{Z})^g} (-1)^{\langle t|v \rangle} \theta_t^B([4]T_i)^2$$

$$\begin{aligned}
&= \frac{\lambda\mu_i}{2^g} \sum_{t \in (\mathbb{Z}/2\mathbb{Z})^g} \left(\sum_{v \in (\mathbb{Z}/2\mathbb{Z})^g} (-1)^{\langle t|v \rangle} \right) \theta_t^A([4]T_i)^2 \\
&= \frac{\lambda\mu_i}{2^g} \sum_{t \in (\mathbb{Z}/2\mathbb{Z})^g} 2^g \delta_{t,0} \theta_t^A([4]T_i)^2 = \lambda\mu_i \theta_0^A([4]T_i)^2.
\end{aligned}$$

This completes the proof. $\square$

Recall the result we want to prove.

Lemma 5. *Assume that we work in the CSURF context, so that $\text{End}_{\mathbb{F}_p}(E) = \mathbb{Z}[(1+\pi)/2]$. Recall the notations from Equation (12) and assume the conditions of Lemma 4 do not apply i.e. that both pairs $\{\sigma_1, \sigma_2\}$ and $\{\sigma_3, \sigma_4\}$ contain an odd element. Then, the codomain $\mathcal{A}_2$ of the second isogeny has one zero theta constant exactly when either one of $\sigma_1, \sigma_2, \sigma_3$ or σ_4 is $2 \pmod{4}$.*

Proof. The strategy to prove Lemma 5 follows naturally from Lemma 6. We compute the theta coordinate $\theta'_0{}^{E^4}([2^e]T_i)$ for all $i \in (\mathbb{Z}/2\mathbb{Z})^4$, where:

$$T_i := \sum_{l=1}^4 [i_l]T_l$$

and T_1, T_2, T_3, T_4 are given by Equation (11) and the theta coordinate $\theta'_0{}^{E^4}$ is associated to the theta structure induced by $[2^e]\mathcal{B}$ also given by Equation (11). To obtain $\theta'_0{}^{E^4}$, we compute a change of theta coordinates from a product theta structure on E^4 . The values we obtain can be computed symbolically and only depend on $N_1, \sigma_1, \sigma_2, \sigma_3$ and σ_4 modulo 4. We can determine when one theta coordinate $\theta'_0{}^{E^4}([2^e]T_i)$ vanishes by enumerating all possible values of $N_1, \sigma_1, \sigma_2, \sigma_3$ and σ_4 modulo 4.

Let (P, Q) be the basis of $E[2^{e+2}]$ such that $\pi(P) = P$ and $\pi(Q) = -Q$ that we introduced in Section 4.1 in the CSURF context. Let $P_4 := [2^e]P$ and $Q_4 := [2^e]Q$. Consider the basis $\mathcal{B}_E := (P_4, Q_4)$ of $E[4]$ and the product symplectic basis of $E[4]$ given by:

$$\begin{aligned}
\mathcal{B}_0 := \mathcal{B}_E^4 = & ((P_4, 0, 0, 0), (0, P_4, 0, 0), (0, 0, P_4, 0), (0, 0, 0, P_4) \\
& (Q_4, 0, 0, 0), (0, Q_4, 0, 0), (0, 0, Q_4, 0), (0, 0, 0, Q_4))
\end{aligned}$$

Then, by Section 4.1 and since $[2^e]T_P = [2^e]T_Q = 0$, the change of basis matrix from $\mathcal{B}_0$ to $[2^e]\mathcal{B}$ is given by:

$$M := \begin{pmatrix} 0 & 0 & 0 & 0 & N_1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & N_1 & 0 & 0 \\ 0 & 0 & \beta\sigma_3 & \beta\sigma_2 & \sigma_3 & \sigma_2 & 0 & 0 \\ 0 & 0 & -\beta\sigma_1 & \beta\sigma_4 & -\sigma_1 & \sigma_4 & 0 & 0 \\ -\alpha & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & -\alpha & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & \alpha\sigma_4 & \alpha\sigma_1 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\alpha\sigma_2 & \alpha\sigma_3 \end{pmatrix}, \quad (15)$$

where $\alpha \equiv N_1^{-1} \pmod{4}$, $\beta \equiv N_2^{-1} \equiv -\alpha \pmod{4}$ and $[2^e]T_1, [2^e]T_2, [2^e]T_3, [2^e]T_4$ are determined by the four last columns of this matrix.

From this data, we can express the theta coordinates $\theta_0^{E^4}([2^e]T_i)$ as follows. We start by computing the product theta coordinates $(\theta_j^{E^4}([2^e]T_i))_j$ associated to $\mathcal{B}_0$ for all $i \in (\mathbb{Z}/2\mathbb{Z})^4$ as:

$$\theta_j^{E^4}([2^e]T_i) = \prod_{l=1}^4 \theta_{jl}^E([2^e]T_{i,l}), \quad (16)$$

where $T_i := (T_{i,1}, T_{i,2}, T_{i,3}, T_{i,4})$ component-wise. To obtain the level 2 theta coordinates $(\theta_0^E : \theta_1^E)$ on E induced by $\mathcal{B}_E = (P_4, Q_4)$ we use the transition formulae from Montgomery $(x : z)$ -coordinates to theta coordinates introduced in [46, Chapter 7, Appendix A]. These formulae apply when $Q_4 = (-1 : * : 1)$.

Recall that because $\text{End}_{\mathbb{F}_p}(E) = \mathbb{Z}[(1+\pi)/2]$ and $p \equiv 7 \pmod{8}$, E can admit two $\mathbb{F}_p$ -isomorphic Montgomery models by [13, Corollary 1]. In Section 4.1, we imposed the Montgomery coefficient A of E to be such that $A+2$ is not a square. Then, $A-2$ is also not a square by [19, Table 1], because $E[2] \subseteq E(\mathbb{F}_p)$. It follows that 4-torsion points of the form $(\pm 1 : * : 1)$ are not $\mathbb{F}_p$ -rational in E but $\mathbb{F}_p$ -rational in E^t . Since $\pi(Q_4) = -Q_4$, it follows that $Q_4 = (\pm 1 : * : 1)$.

Assume that $Q_4 = (-1 : * : 1)$. Then the transition formulae apply. The theta null point of E associated to $\mathcal{B}_E = (P_4, Q_4)$ is of the form $(a : b) := (x(P_4) + z(P_4) : x(P_4) - z(P_4))$, so we can write $P_4 := (a + b : * : a - b)$. We also have:

$$(\theta_0^E : \theta_1^E) = (a(x - z) : b(x + z)), \quad (17)$$

where $(x : z)$ are the Montgomery coordinates of E . It follows that $(\theta_0^E(P_4) : \theta_1^E(P_4)) = (1 : 1)$ and $(\theta_0^E(Q_4) : \theta_1^E(Q_4)) = (1 : 0)$. By [22, Eq. (5.12)], we also obtain $(\theta_0^E([2]P_4) : \theta_1^E([2]P_4)) = (b : a)$, $(\theta_0^E([2]Q_4) : \theta_1^E([2]Q_4)) = (a : -b)$ and $(\theta_0^E([2](P_4 - Q_4)) : \theta_1^E([2](P_4 - Q_4))) = (b : -a)$. Using the point duplication algorithm [22, Algorithm 5.2] in dimension 1, to solve the equation

$$[2](\theta_0^E(P_4 - Q_4) : \theta_1^E(P_4 - Q_4)) = (b : -a),$$

we see that $(\theta_0^E(P_4 - Q_4) : \theta_1^E(P_4 - Q_4)) = (1 : \zeta_4)$, with $\zeta_4^2 = -1$. Using the differential addition algorithm [22, Algorithm 5.1] in dimension 1, we can precompute the linear combinations $(\theta_0^E([u]P_4 + [v]Q_4) : \theta_1^E([u]P_4 + [v]Q_4))$ in the following table:

Now, if $Q_4 = (1 : * : 1)$, then we have $Q_4 = Q'_4 + [2]P_4$ with $Q'_4 := (-1 : * : 1)$ so $\mathcal{B}_E$ is obtained from $\mathcal{B}'_E := (P_4, Q'_4)$ via the change of basis matrix (in columns):

$$\begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}.$$

Applying [22, Theorem 6.2.10] to the above matrix and Equation (17), we obtain that the theta structure induced by $\mathcal{B}_E$ satisfies:

$$(\theta_0^E : \theta_1^E) = (b(x + z) : a(x - z)),$$

$v \backslash u$	0	1	2	3
0	$(a : b)$	$(1 : 0)$	$(a : -b)$	$(1 : 0)$
1	$(1 : 1)$	$(1 : -\zeta_4)$	$(1 : -1)$	$(1 : \zeta_4)$
2	$(b : a)$	$(0 : 1)$	$(b : -a)$	$(0 : 1)$
3	$(1 : 1)$	$(1 : \zeta_4)$	$(1 : -1)$	$(1 : -\zeta_4)$

Table 3. Values $(\theta_0^E([u]P_4 + [v]Q_4) : \theta_1^E([u]P_4 + [v]Q_4))$ for $u, v \in \mathbb{Z}/4\mathbb{Z}$.

where $P_4 = (a + b : * : a - b)$. It follows that $(\theta_0^E(P_4) : \theta_1^E(P_4)) = (1 : 1)$ and $(\theta_0^E(Q_4) : \theta_1^E(Q_4)) = (1 : 0)$ again and that the theta null point is $(\theta_0^E(\mathbf{0}) : \theta_1^E(\mathbf{0})) = (b : a)$. We then obtain the linear combinations $(\theta_0^E([u]P_4 + [v]Q_4) : \theta_1^E([u]P_4 + [v]Q_4))$ from Table 3 but with b and a reversed. Symbolically, the computations are strictly the same so we can swap a and b back or assume that $Q_4 = (-1 : * : 1)$ without loss of generality.

Expressing the $[2^e]T_i$ component-wise in terms of linear combinations of P_4 and Q_4 and combining Equation (16) with Table 3, we can express their product theta coordinates. Then applying [22, Theorem 6.2.10] to the matrix M from Equation (15), we obtain the $\theta_0'^{E_4}([2^e]T_i)$, as desired. Doing the computations symbolically in SageMath⁶ and enumerating all possible values of $N_1, \sigma_1, \sigma_2, \sigma_3$ and σ_4 modulo 4 that do not satisfy the condition of Lemma 4 and such that M is a symplectic matrix, we obtain that $\theta_0'^{E_4}([2^e]T_i) = 0$ for either $i = (0, 1, 1, 0)$ or $i = (1, 0, 0, 1)$ when one of the values $\sigma_1, \sigma_2, \sigma_3$ or σ_4 is $2 \pmod 4$. This does not happen otherwise. This completes the proof. $\square$

⁶Our symbolic computation code can be found in the qt-Pegasis implementation public repository <https://github.com/KULeuven-COSIC/qt-pegasis> in the file `experiments/singularities.py`